

Final report on
Art Gallery Management System

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Course Number & Name:

CSE 3224 - Information & System Design Lab

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Project Overview

The Art Gallery Management System is a comprehensive software solution designed to streamline and automate art gallery operations. It integrates key functionalities such as artist and artwork management, inventory tracking, sales and transactions, event organization, visitor browsing and reporting. This system enhances efficiency, accuracy and customer experience by providing a cohesive platform for managing artwork inventory, artist profiles and customer information. With features like submission processes, sales tracking and reporting analytics, the system empowers gallery owners to showcase artwork effectively, provide personalized service, and make informed decisions. The Gallery Management System optimizes operations, elevates the visitor experience, and ensures smooth management of art galleries in a user-friendly manner.

Project Implementation Tools/Framework

- Netbeans 8.2
- Java Swing
- JDBC Driver: sqljdbc 4.0
- MSsql
- SQL Server 2014 Management Studio

User Story:

1. As a gallery employee, I want to enter new artwork into the Gallery Management System, categorizing them according to different art forms and genres so that they can be easily managed and organized for efficient inventory management.

- The system should provide an interface to input new artwork, update existing artwork details and delete artwork if necessary.
- I should be able to specify details such as the title, artist name, medium, dimensions and price of the artwork.
- The system should allow me to assign the artwork to relevant categories or collections, such as paintings, sculptures, photography etc.

2. As a gallery employee, I want to input customer information into the Gallery Management System, so that we can provide personalized service and maintain an up-to-date customer database.

- I should be able to input customer details, including name, contact information, address and any relevant preferences or notes.
- The system should securely store the customer information for future interactions and transactions.
- I should be able to update and delete customer profiles as necessary.
- Upon successful update, the changes should be reflected in the customer's profile for future interactions and transactions.

3. As a gallery employee, I want to generate invoices for artwork purchases using the Gallery Management System, applying appropriate details based on the customer's purchases and any applicable discounts or promotions.

- The system should calculate the total price of purchased artwork, including any applicable taxes or fees.
- It should apply appropriate discounts or promotions based on the customer's profile or ongoing promotions.
- The system should provide options to print or email the invoice to the customer.

4. As a gallery employee, I want to update artwork details in the Gallery Management System, so that accurate and up-to-date information is available to customers.

- The system should allow me to update artwork details such as title, artist, medium, dimensions, and price.
- It should provide a simple interface to make changes and save the updated information.

Module Breakdown

1.Login Administration:

- Admin login: Gives admin access to the entire system and the ability to create accounts, log in, and access them.
- User login: Users can log in to the database at any time by creating an account with a username and password.

2. Employee Directory:

- Employee List: Shows a thorough list of every employee.
- Employee Details: Displays specific details about each.

3. Management of Artists List:

- Artists list management: This feature enables the addition, deletion, and updating of artist's data in the artists list.
- Searching Artists: This feature enables you to look for individual artists using a variety of parameters.

4. Management of Painting Information:

- Add painting: Enables store staff to insert information like the description, the date, and the prices appropriately.
- Categories: Different genres are used to categorize paintings.

- Size: Paintings are organized based on their dimensions, size and design.
- Painting information: Names the artist(s) and their work.
- Erase painting: In accordance with shop standards, employees may erase any information about the paintings.

5. Customer Information Management:

- Add Customer: Allows the input of painting information such as name, contact information (phone, email, etc.) and membership availability.
- Update Customer: Allows store employees to update the Customer details.
- Customer Information: Each customer who has an entry in the database is accessible to store workers.
- Delete Customer: Depending on the availability, store staff may delete any customer information.

6. Invoice generation:

- Store employees can search any product or customer details using only one unique key for the Invoice.
- Based on the customer's choices, computes the total cost of the purchased paintings, including any applicable vats.
- Offers discounts in accordance with the customer' status as indicated by their profile.
- Provides choices for clients to print invoices.
 - Generates detailed invoices that include the date, time, list of paintings, pricing, and discounts.

7.Daily Sales Management:

- Displays all information about the paintings being sold together with the customers purchasing times.

8.Log out and System Termination:

- Log Out: Displays the login interface and enables users to log out of the application.
- Terminate the application: This option is available to the store workers.

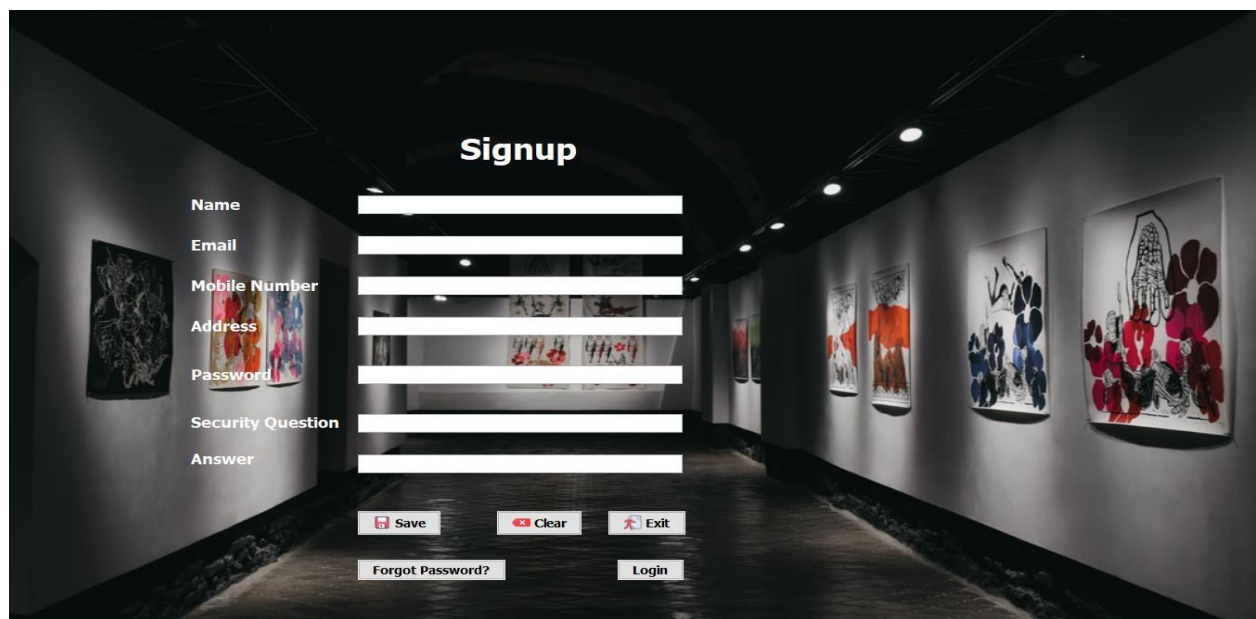
9.Bidding Management:

- Bidding Setup: Administrators and authorized store workers will have the ability to configure auctions by specifying parameters such as auction duration, starting bid price, bid increments, reserve prices, and maximum bid amounts.
- Artwork Selection: Store workers will be able to select artworks from the gallery inventory to include in the bidding process, defining the start and end dates for each auction.

Prototype Design:

User Stories:

As a gallery employee, I want users to be able to create a new account by providing their name, email, phone number, address, password and security question answer. This will allow them to access the system's features and ensure the security of my employee account. So we created Fig-(1).

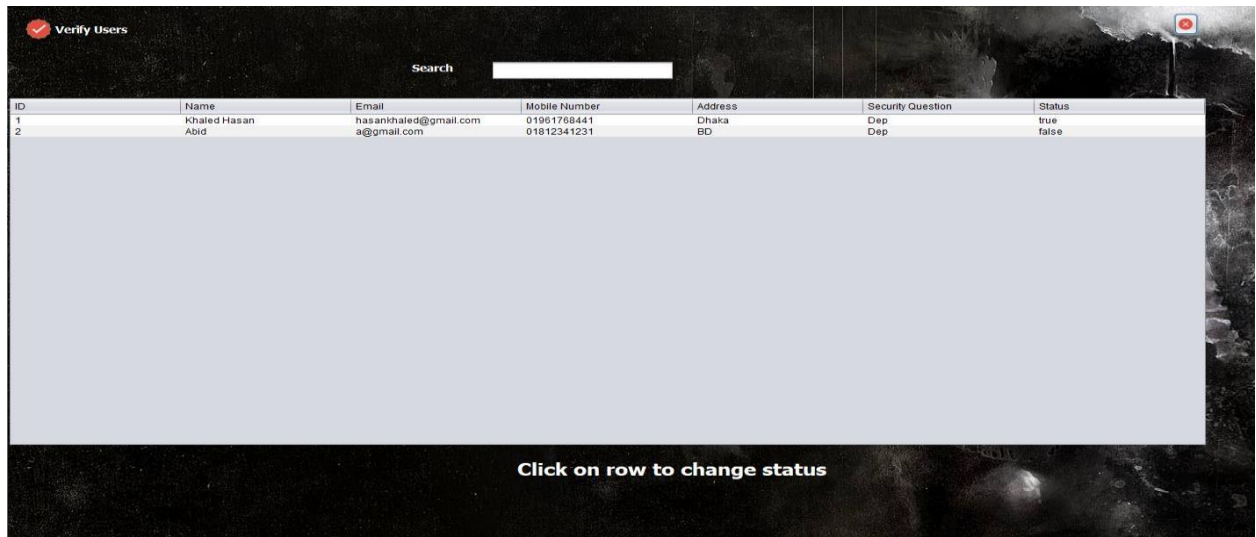


The image shows a 'Signup' form overlaid on a background of a gallery hallway with art on the walls. The form is titled 'Signup' and contains the following fields and buttons:

- Name**: Input field
- Email**: Input field
- Mobile Number**: Input field
- Address**: Input field
- Password**: Input field
- Security Question**: Input field
- Answer**: Input field
- Buttons**: 'Save', 'Clear', 'Exit', 'Forgot Password?', and 'Login'.

Fig-(1): Signup UI

As a gallery employee, I want to verify the login information provided by users to ensure secure access to the system. This verification process will help prevent unauthorized access and protect sensitive user data. So we created Fig-(2).

The image shows a web application interface titled "Verify Users". It features a search bar at the top. Below it is a table with columns: ID, Name, Email, Mobile Number, Address, Security Question, and Status. The table contains two rows of data. Below the table is a large grey rectangular area. At the bottom, there is a text prompt: "Click on row to change status".

ID	Name	Email	Mobile Number	Address	Security Question	Status
1	Khaled Hasan	hasankhaled@gmail.com	01951768441	Dhaka	Dep	true
2	Abid	a@gmail.com	01812341231	BD	Dep	false

Fig-(2): Verify Users UI

As a gallery employee, I want to log in to the Art Gallery Management System so that I can access the necessary features and perform my duties effectively. I should be successfully logged into the system using the user email and password. So we created Fig-(3).



Fig-(3): Login UI

As a gallery employee, I want the users have the ability to recover their forgotten password by answering a security question associated with their email address. This will provide a secure and convenient method to regain access to their account and ensure the confidentiality of my work-related information. So we created Fig-(4).

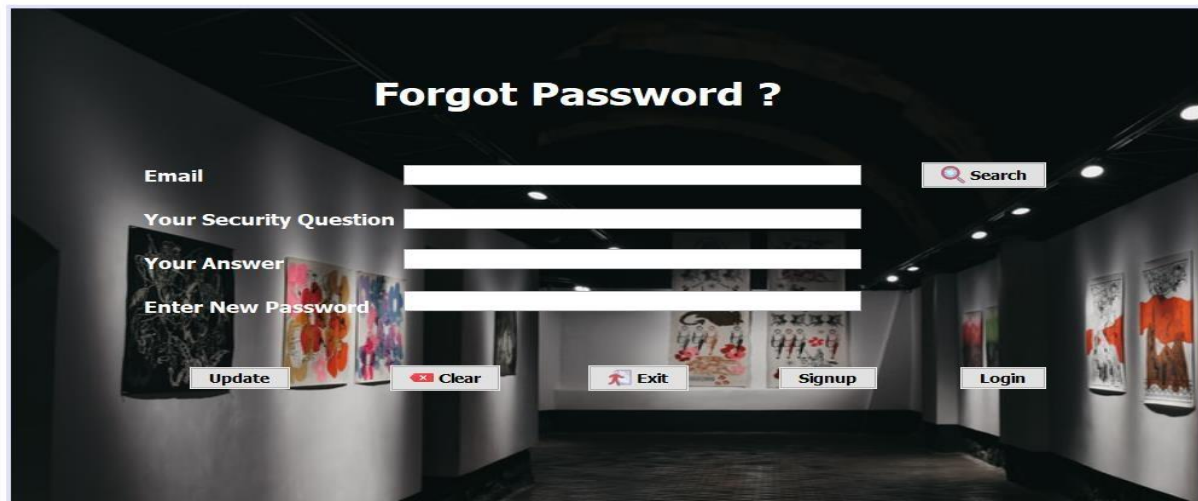
The image shows a 'Forgot Password ?' interface overlaid on a gallery background. The interface includes four input fields for 'Email', 'Your Security Question', 'Your Answer', and 'Enter New Password'. Below these fields are buttons for 'Update', 'Clear', 'Exit', 'Signup', and 'Login'. A search bar with a magnifying glass icon and the word 'Search' is located in the top right corner.

Fig-(4): Forgot Password UI

As a gallery employee, I want to have a dedicated admin home page to access important administrative functionalities and monitor the system. Additionally, as a user, I want to have a user home page that provides a personalized experience and easy access to relevant features and information. So we created Fig-(5)&Fig-(6).

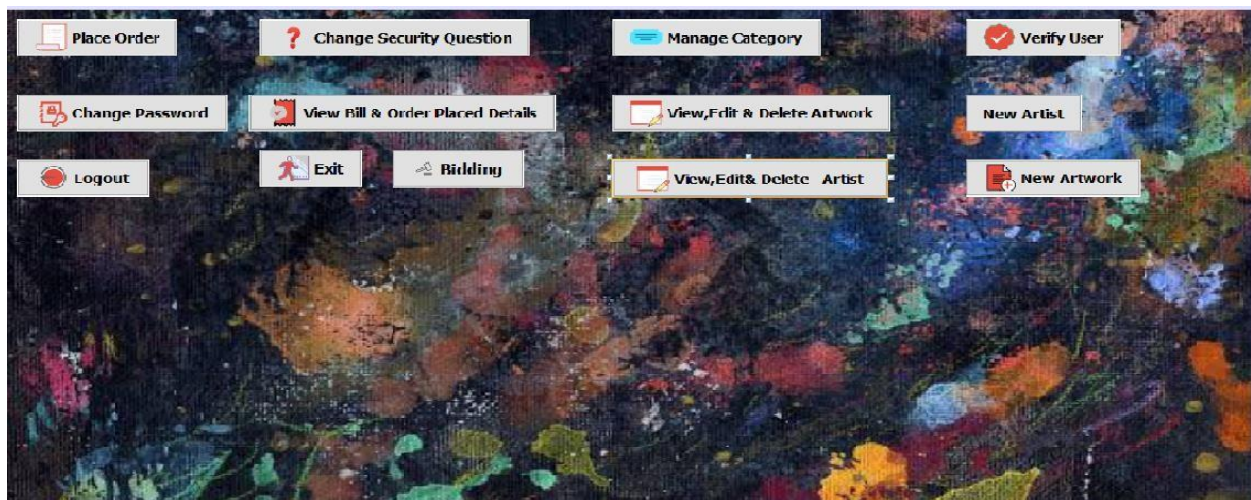


Fig-(5): Admin Home page UI



Fig-(6): User Home page UI

As a gallery employee, I want to be able to manage art categories effectively to ensure proper organization and easy navigation for users browsing artworks. This includes creating, updating, and deleting art categories as needed. So we created Fig-(7).

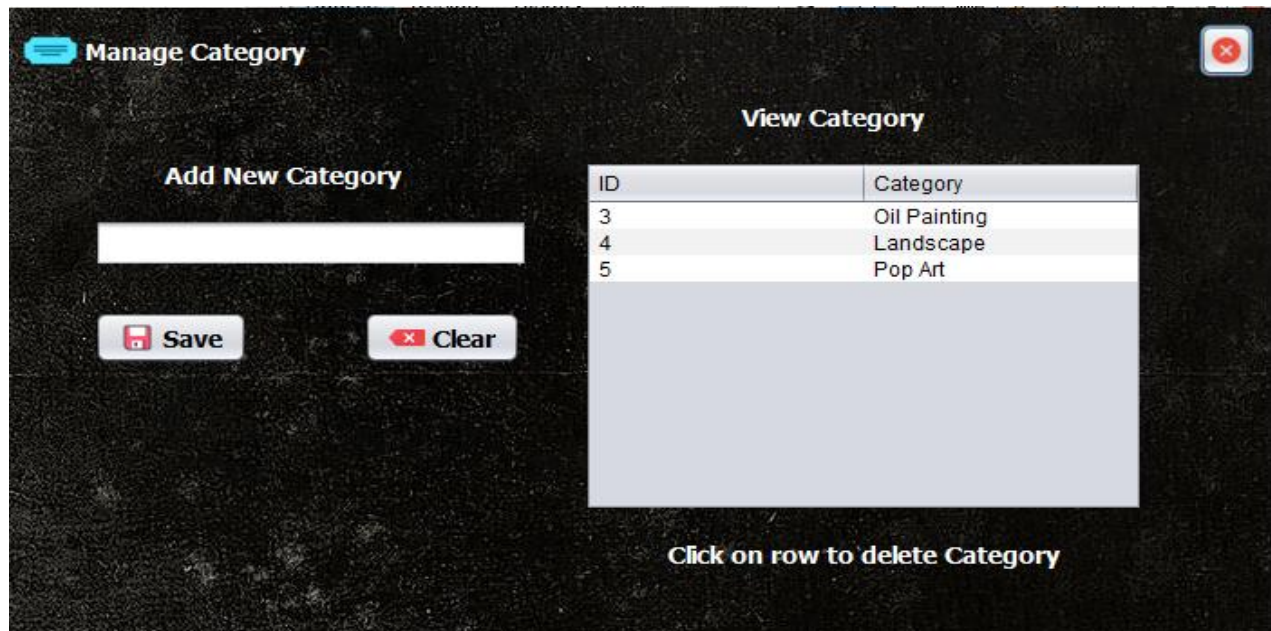


Fig-(7): Manage Category UI

As a gallery employee, I want to enter new artwork into the gallery management system categorizing them according to different art forms and genres so that they can be easily managed and organized for efficient inventory management. So we created Fig-(8).

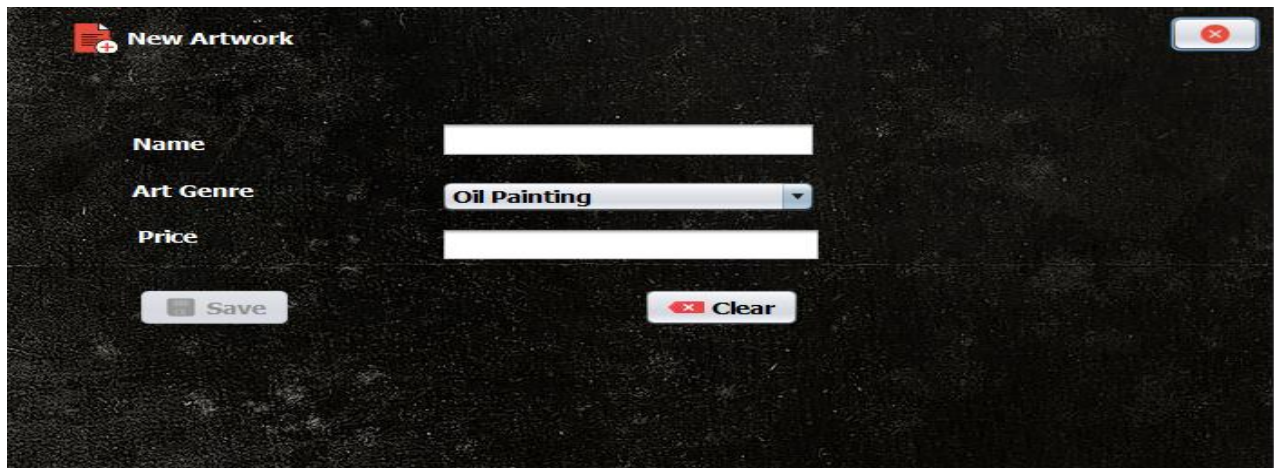
The image shows a software interface titled "New Artwork" with a red document icon and a close button in the top right. It features three input fields: "Name" (a text box), "Art Genre" (a dropdown menu currently showing "Oil Painting"), and "Price" (a text box). At the bottom, there are two buttons: "Save" with a floppy disk icon and "Clear" with a red 'x' icon.

Fig-(8): New Artwork UI

As a gallery employee, I want to input customer information into the gallery management system so that we can provide personalized service and maintain an up-to-date customer database. So we created Fig-(9).

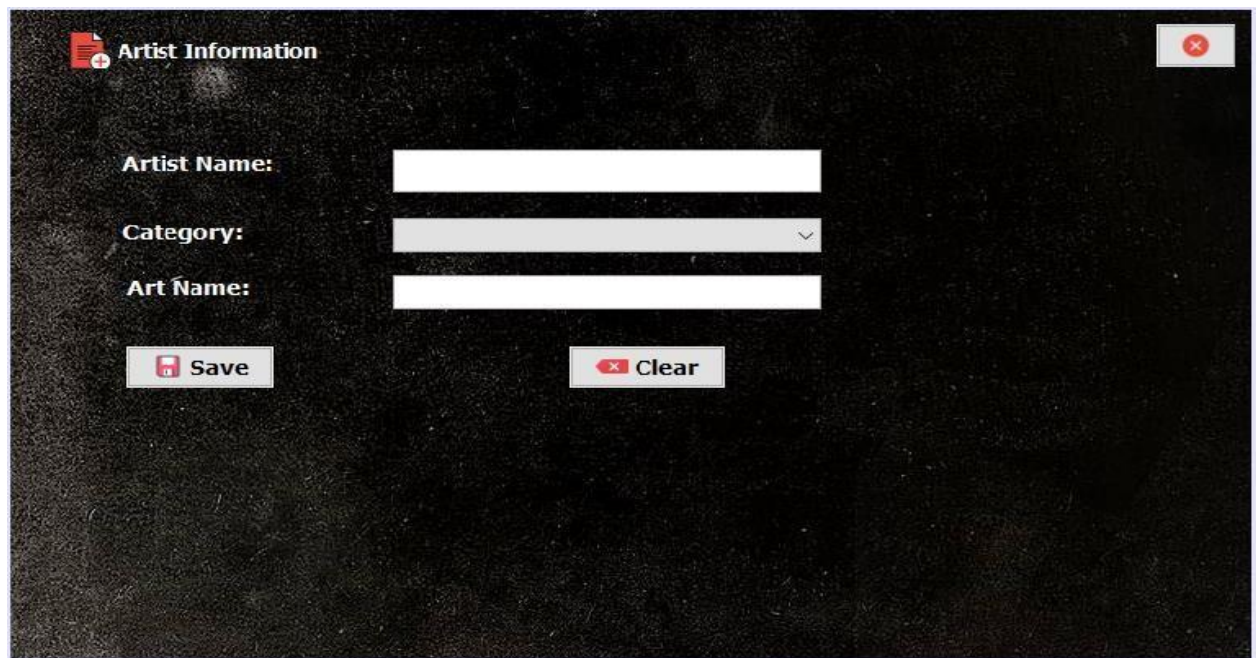
The image shows a software interface titled "Artist Information" with a red document icon and a close button in the top right. It features three input fields: "Artist Name:" (a text box), "Category:" (a dropdown menu), and "Art Name:" (a text box). At the bottom, there are two buttons: "Save" with a floppy disk icon and "Clear" with a red 'x' icon.

Fig-(9): Add Artist Information UI

As a **gallery employee**, I must manage the information about artists. I need to change, delete or examine the artist table. They can also add a table with titles of any artists' pieces. So we created Fig-(10).

View, Edit, Delete of Artist

ID: 00

Artist Name:

Category:

Art Name:

ID	Artist Name	Category	Art Name
1	Leonardo da ...	Oil Painting	Mona Lisa

Fig-(10): View Artist UI

As a **gallery employee**, I have to organize all the painting information as I need to place the artwork on the table, change it if needed or remove a few paintings from it when required. Additionally, we classify the artwork and set prices for it. So we created Fig-(11).

View ,Edit,Delete painting

ID: 00

Art Name:

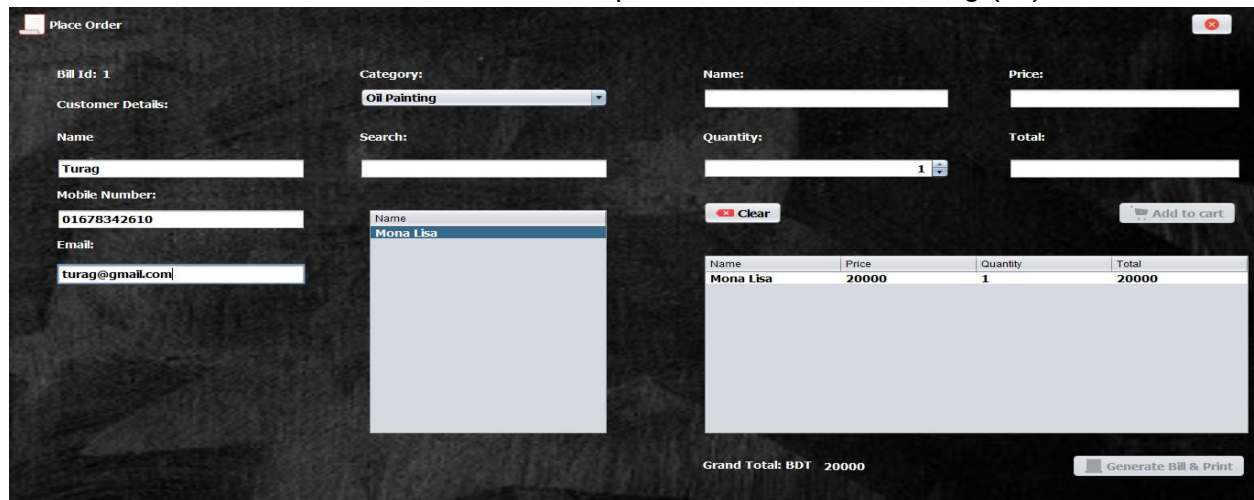
Category:

Price:

ID	Name	Category	Price
2	Mona Lisa	Oil Painting	20000

Fig-(11): View Painting UI

As a gallery employee, I want users to be able to place an order by providing their details and selecting products including their names, prices, quantities and total prices, to be added to a cart for a seamless and convenient checkout experience. So we created Fig-(12).



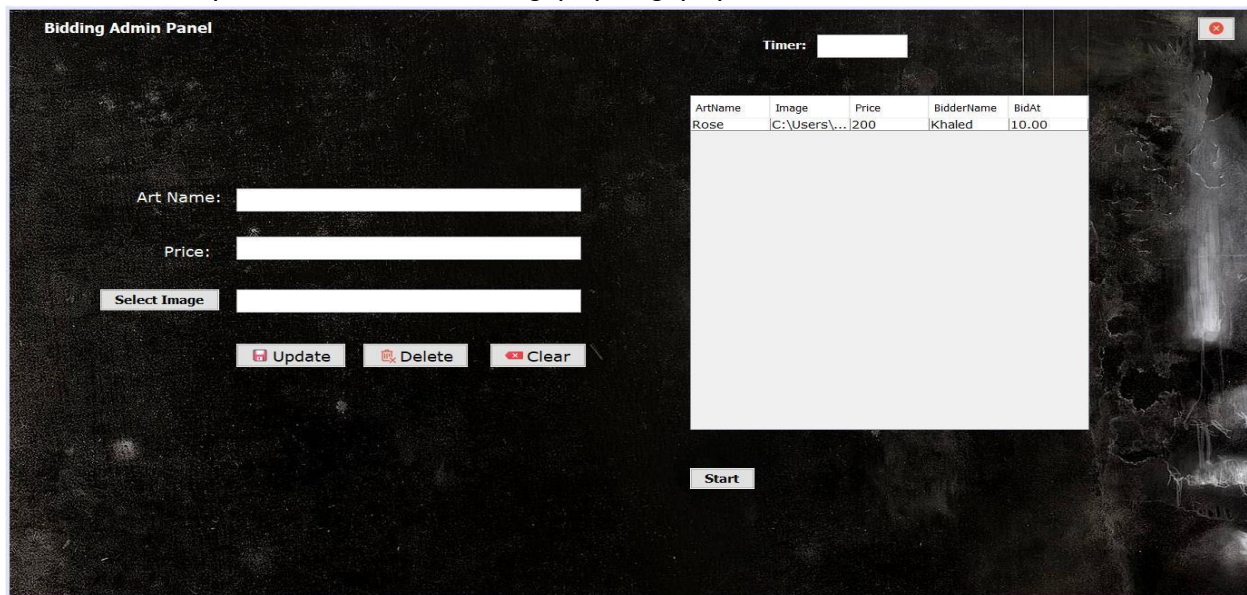
The screenshot shows a 'Place Order' form with the following sections:

- Customer Details:**
 - Bill Id: 1
 - Name: Turag
 - Mobile Number: 01678342610
 - Email: turag@gmail.com
- Category:** Oil Painting
- Search:** A search bar with a dropdown menu showing 'Mona Lisa'.
- Product Selection:**
 - Name: [Empty]
 - Price: [Empty]
 - Quantity: 1
 - Total: [Empty]
- Buttons:** Clear, Add to cart
- Table:**

Name	Price	Quantity	Total
Mona Lisa	20000	1	20000
- Grand Total:** BDT 20000
- Buttons:** Generate Bill & Print

Fig-(12): Place Order UI

As an employee, I want to have an admin panel and a customer panel to manage and facilitate the bidding process. The admin panel should allow me to update, delete, and clear bid listings while the customer panel should enable customers to view available artwork and place bids with their desired bid prices. So we created Fig-(13)&Fig-(14).



The screenshot shows a 'Bidding Admin Panel' with the following sections:

- Form Fields:**
 - Art Name: [Empty]
 - Price: [Empty]
 - Select Image: [Empty]
- Buttons:** Update, Delete, Clear
- Table:**

Artname	Image	Price	BidderName	BidAt
Rose	C:\Users\... \200	200	Khaled	10.00
- Buttons:** Start

Fig-(13): Bidding Admin Panel UI

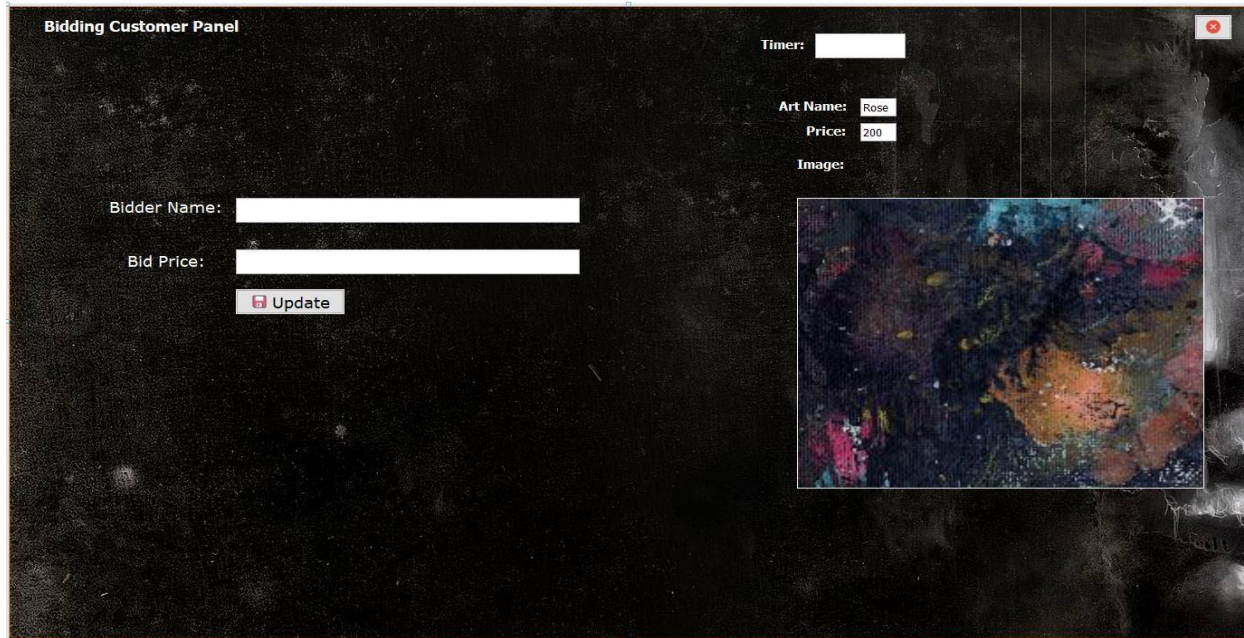


Fig-(14): Bidding Customer Panel UI

To terminate the system the exit button needs to be pressed from the Fig:- (15), which will close the whole system.

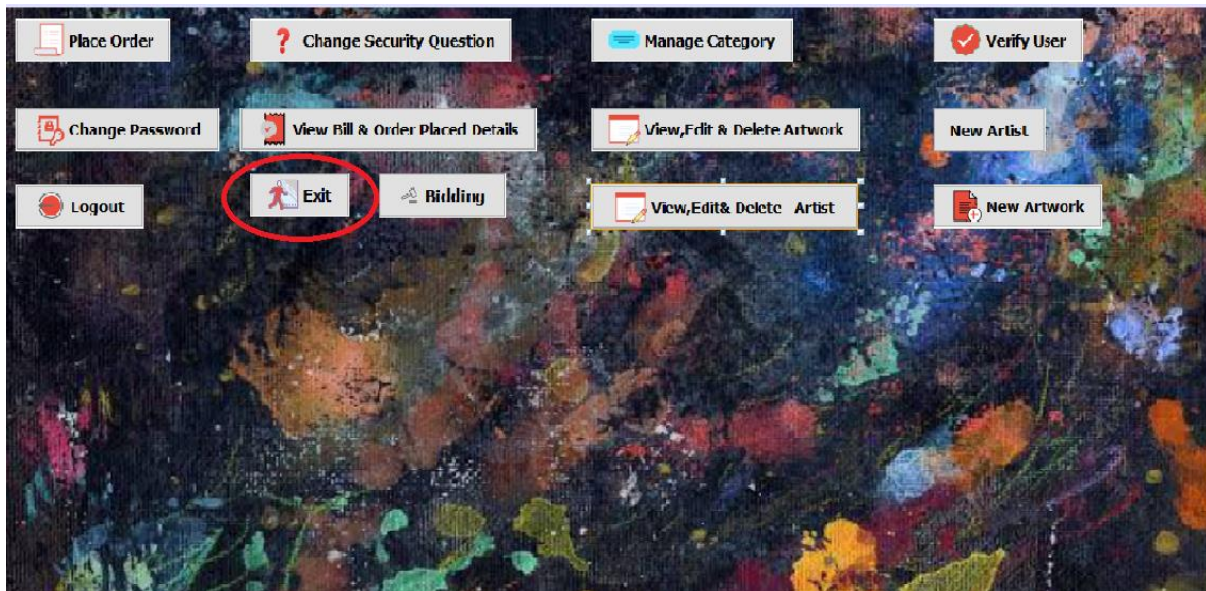


Fig-(15): Admin Home page UI

Functional Points Analysis

Step 1: Function point Ranking According to Complexity

Table 1

Measurement Parameter	Low	Average	High
Number of external inputs (EI)	7	10	15
Number of external outputs (EO)	5	7	10
Number of external inquiries (EQ)	3	4	6
Number of internal files (ILF)	4	5	7
Number of external interfaces (EIF)	3	4	6

Step 2: Calculate Unadjusted Function Point (UFP)

Estimate the Complexity Values for each Functional Point Components:

External Inputs (EI):

- Signup: 1 EI (Low) - Basic authentication (username/password check)
 - Verify Users: 1 EI (Low) - Simple user verification process
-

- Admin Login: 1 EI (Low) - Basic admin authentication
- User Login: 1 EI (Low) - Basic user authentication
- Forgot Password: 1 EI (Low) - Password recovery process
- Manage Category: 1 EI (Low) - Basic category management
- Add New Art: 1 EI (Low) - Simple data entry for new art
- Update Art: 1 EI (Low) - Basic art information update
- Delete Art: 1 EI (Low) - Basic art deletion
- Add New Artist: 1 EI (Low) - Simple data entry for new artist
- Update Artist: 1 EI (Low) - Basic artist information update
- Delete Artist: 1 EI (Low) - Basic artist deletion
- Place Order: 1 EI (High) - Complex process of placing an order
- Bidding Admin Panel: 1 EI (High) - Complex bidding admin panel interaction

Total Weighing Factor: $7 * 12 + 10 * 0 + 15 * 2 = 114$

External Outputs (EO):

- User Home Page: 1 EO (Low) - Displaying user's home page
- View Art: 1 EO (Low) - Displaying art details
- View Artist: 1 EO (Low) - Displaying artist details
- View Bill: 1 EO (Low) - Displaying bill information
- Bidding User Panel: 1 EO (High) - Complex bidding user panel interaction

Total Weighing Factor: $5 * 4 + 10 * 1 = 30$

External Inquiries (EQ):

- View Art: 1 EQ (Low) - Retrieving and displaying art details
- View Artist: 1 EQ (Low) - Retrieving and displaying artist details
- View Bill: 1 EQ (Low) - Retrieving and displaying bill information

Total Weighting Factor: $3 \times 3 = 9$

Internal Files (ILF):

- Manage Category: 1 ILF (Average) - Storing category management data.
- Add New Art: 1 ILF (Low) - Storing information about newly added artworks.
- Update Art: 1 ILF (Low) - Storing updated information for artworks.
- Delete Art: 1 ILF (Low) - Storing data related to deleted artworks.
- Add New Artist: 1 ILF (Low) - Storing details of newly added artists.
- Update Artist: 1 ILF (Low) - Storing updated information for artists.
- Delete Artist: 1 ILF (Low) - Storing data related to deleted artists.
- Place Order: 1 ILF (Average) - Storing order placement details.

Total Weighting Factor: $4 \times 6 + 5 \times 2 + 7 \times 0 = 34$

External Interface Files (EIF):

Total Weighting Factor: $4 \times 0 = 0$

Table 2

Measurement Parameter	Weighing Factor
EI	114
EO	30
EQ	9
ILF	34
EIF	0
Total Count	187

So, **UFP** = 187

Step 3: Calculating Complexity Adjustment Factor (CAF)

Table 3

General System Characteristics	Degree Of Influence (DI)
1.Data Communications	5
2.Distributed Data Processing	0
3.Performance	3
4.Heavily User Configuration	2
5.Transaction Rate	2
6.Online Data Entry	3
7.End-User Efficiency	5
8.Online Update	0
9.Complex Processing	5
10.Reusability	2
11.Installation Ease	4
12.Operational Ease	5
13.Multiple Sites	0
14.Facilitate Change	3
TDI	39

0-Not present; 1-Insignificant; 2-Moderate; 3-Avarage; 4-Significant; 5-Essential

So, **CAF** = $[0.65 + (0.01 * \mathbf{TDI})] = [0.65 + (0.01 * 39)] = 1.04$

Step 4: Calculating Functional Points (FP)

So, **FP** = **UFP** * **CAF** = $195 * 1.04 = 194.48$

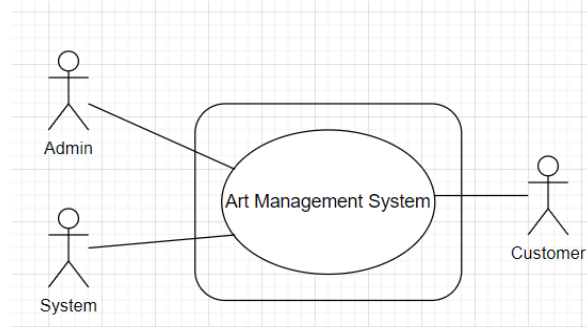
Gantt Chart

The Gantt chart presented below provides a visual representation of the project schedule for the Development of our Art Gallery Management System.

Task	Start Date	Days To Complete
1 Planning & Design Phase	2-May	10
1.1 Project Planning & Features Selection	2-May	4
1.2 Create a Roadmap	6-May	3
1.3 Create a Roadmap	9-May	3
2 Development Phase	23-May	45
2.1 Login & SignUp Design & Implementation	23-May	3
2.2 Home Page UI Design	26-May	9
2.3 Database Connection & Design	26-May	14
2.4 Backend Implementation of Manage Category	9-Jun	5
2.5 Backend Implementation of Place Order	14-Jun	5
2.6 Backend Implementation of View Edit & Delete	19-Jun	5
2.7 Backend Implementation of View Edit & Delete Artist	24-Jun	5
2.8 Backend Implementation of Forgot Password	27-Jun	3
2.9 Backend Implementation of Invoice	2-Jul	5
2.10 Backend Implementation of Bidding	2-Jul	5
3 Testing & Quality Assurance	13-Jul	14
3.1 Bug Finding & Solving	13-Jul	7
3.2 Device Testing	20-Jul	4
3.3 Process Testing	24-Jul	3
4 Launch	27-Jul	10
4.1 Report Writing	27-Jul	7
4.1 Final Launch	3-Aug	3

Use case diagrams:

LEVEL-0 USE CASE - AGMS



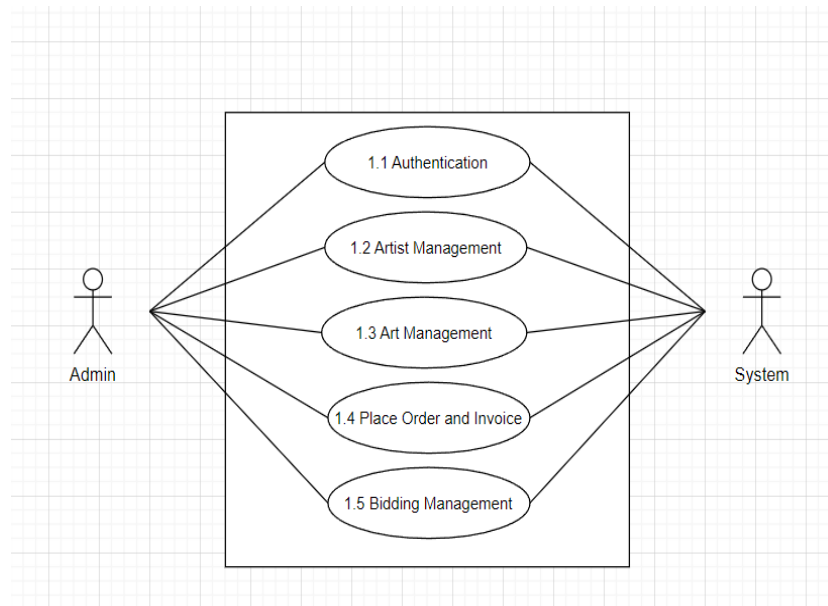
Level-0 Use case-AGMS

Description of Use case diagram level-0:

There are 2 actors who directly or indirectly interact with the system. Primary actors are those who will play action and get a reply from the system whereas secondary actors only produce or consume information. The actors are -

1. Administrator or System
2. Customer

LEVEL-1 USE CASE – Sub system



Level-1 Use case-Sub system

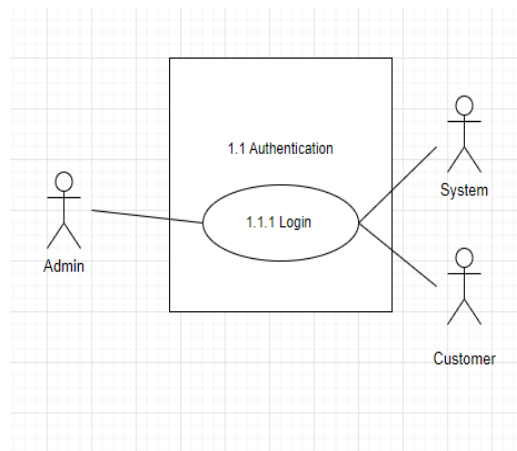
Description of Use case diagram level-1:

There are five subsystems in the Art Gallery Management System. They are :

1. Authentication
2. Artist management
3. Art management
4. Place order and invoice
5. Bidding Management

The five subsystems are decomposed, in levels 1.1, 1.2, 1.3, 1.4 and 1.5 respectively.

LEVEL-1.1 USE CASE - Authentication



Level-1.1 Use case-Authentication

Description of Use case diagram level-1.1:

When the user wants to login, he needs to enter the usermail and password. If the usermail and password match, then, the login is successful "Login successful" message is shown. The system shows an error message if the usermail or password, or both are wrong and the user can try again to log into the system.

Action Reply:

User (Administrator+Customer):

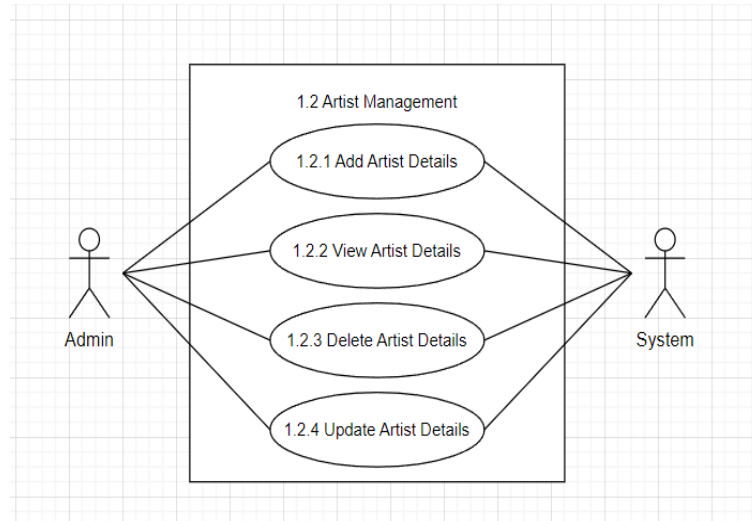
A1: User provides usermail and password

R1: System checks validity. If the usermail and password is valid, the system will allow the user to log into the account.

A2: User provides invalid information. `

R2: The system will show an error message and allow the user to try again.

LEVEL-1.2 USE CASE - Artist management



Level-1.2 Use case-Artist management

Description of Use case diagram level-1.2:

The admin can add new artists to the database, delete and update artists details from the database as well as can also see the details of all the artists from the database.

Action Reply:

User (Administrator):

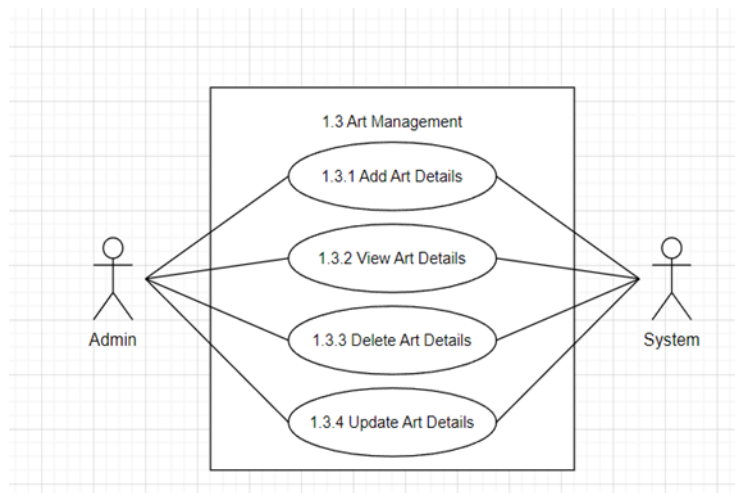
A1: User selects add new artists, delete artists, updates artists details options.

R1: The system acts accordingly and saves the changes.

A2: User selects the view artists details option.

R2: The system will show all the details of all the artists from the database.

LEVEL-1.3 USE CASE - Art management



Level-1.3 Use case-Art management

Description of Use case diagram level-1.3:

The admin can add new art to the inventories, delete and update art details from the inventories as well as can also see the details of all the art present in the inventories.

Action Reply:

User (Administrator):

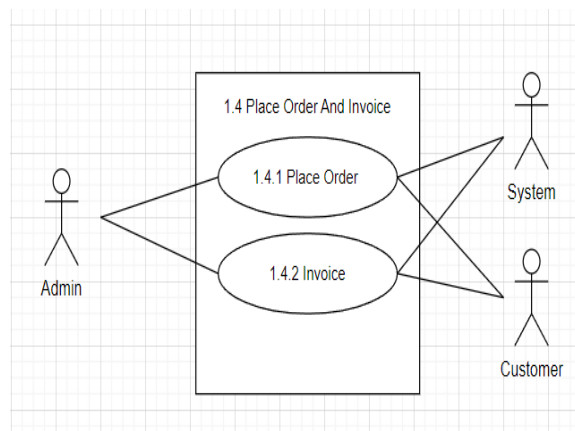
A1: User selects add new arts, delete arts, update arts details options.

R1: The system acts accordingly and saves the changes.

A2: User selects the view art details option.

R2: The system will show all the details of the available arts in the inventories.

LEVEL-1.4 USE CASE – Place order and invoice



Level-1.4 Place order and invoice

Description of Use case diagram level-1.4:

The user can see the details of the place order as well as will be able to generate invoices for the customers.

Action Reply:

User (Administrator+Customer):

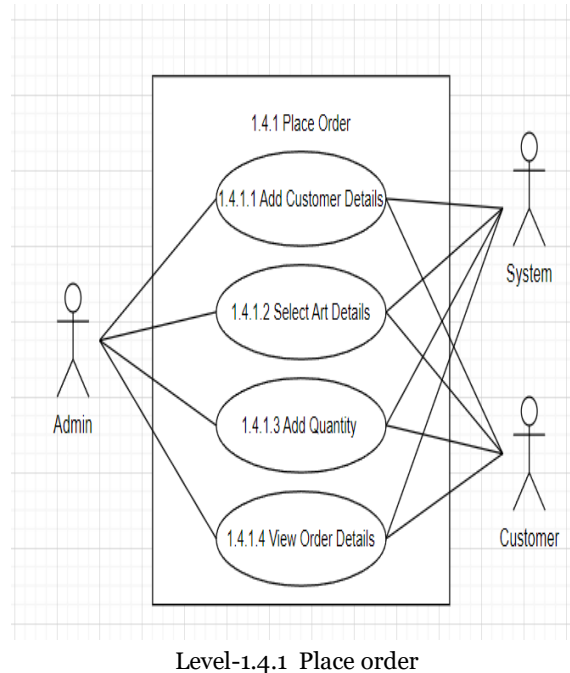
A1: User selects place order option for order.

R1: The system acts accordingly and shows the details.

A2: User selects the invoice generation.

R2: The system will generate a PDF of the invoice for the customer.

LEVEL-1.4.1 USE CASE – Place order



Description of Use case diagram level-1.4.1:

The user can add customer details, select art details, add quantity of art, see the order details. Customer can also add customer details, select art details, add quantity of art, see the order details.

Action Reply:

User (Administrator+Customer):

A1: User selects add customer details, select art details options.

R1: The system acts accordingly.

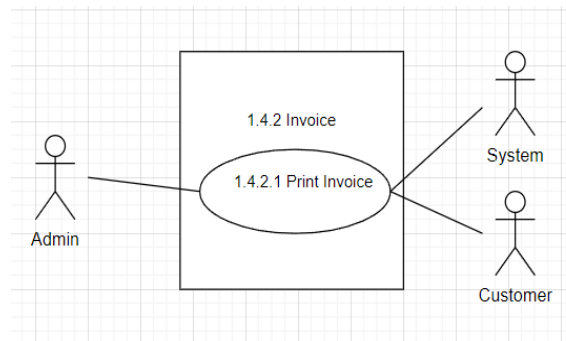
A2: User selects the view order details option.

R2: The system will show all the order details .

A3: User selects add quantity option.

R3: The system acts accordingly and saves the changes.

LEVEL-1.4.2 USE CASE - Invoice generation



Level-1.4.2 Use case-Invoice generation

Description of Use case diagram level-1.4.2:

The user can get the details of the invoice. PDF of the invoice will be generated for the customers.

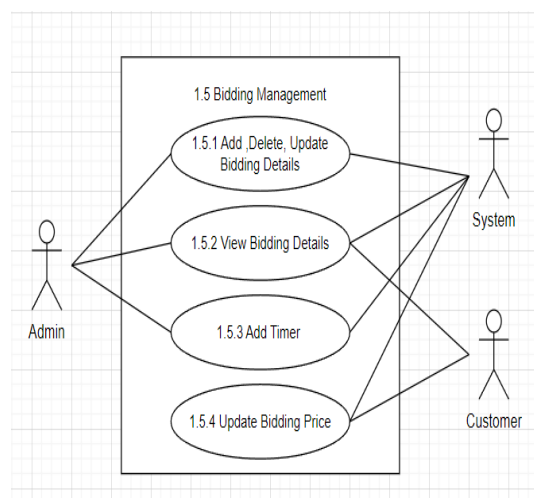
Action Reply:

User (Administrator+Customer):

A1: The user saves the invoice after generating all the information after each sale.

R1: The system will automatically generate a PDF of the invoice for the customer.

LEVEL-1.5 USE CASE – Bidding Management



Level-1.5 Use case-Bidding Management

Description of Use case diagram level-1.5:

The user can add new art bidding details, delete and update bidding details.see the details of bidding details,add timer and add bidding price.Customer can see the details of bidding details, add bidding price.

Action Reply:

User (Administrator/Customer):

A1: User (admin) selects add bidding details, delete bidding details, updates a bidding details options.

R1: The system acts accordingly and saves the changes.

A2: User (admin & customer) selects the view bidding details option.

R2: The system will show all the bidding details .

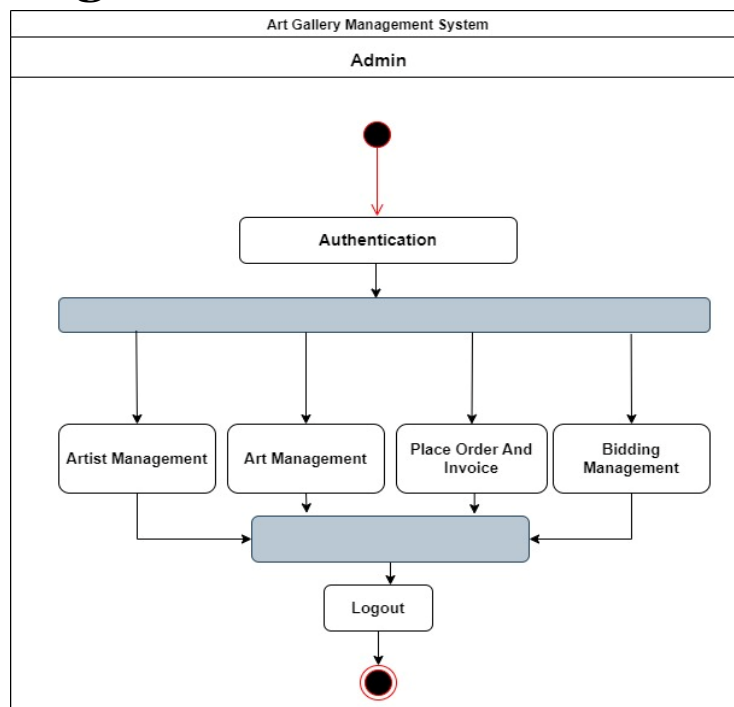
A3: User(admin) selects add timer option.

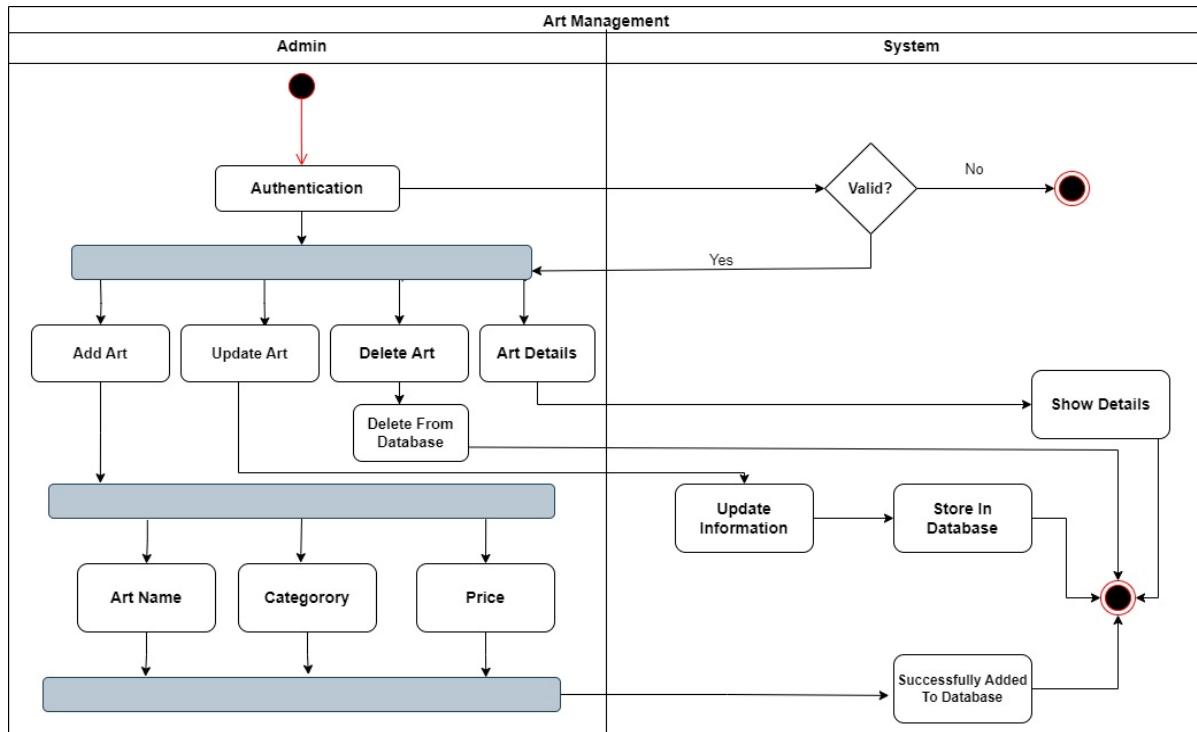
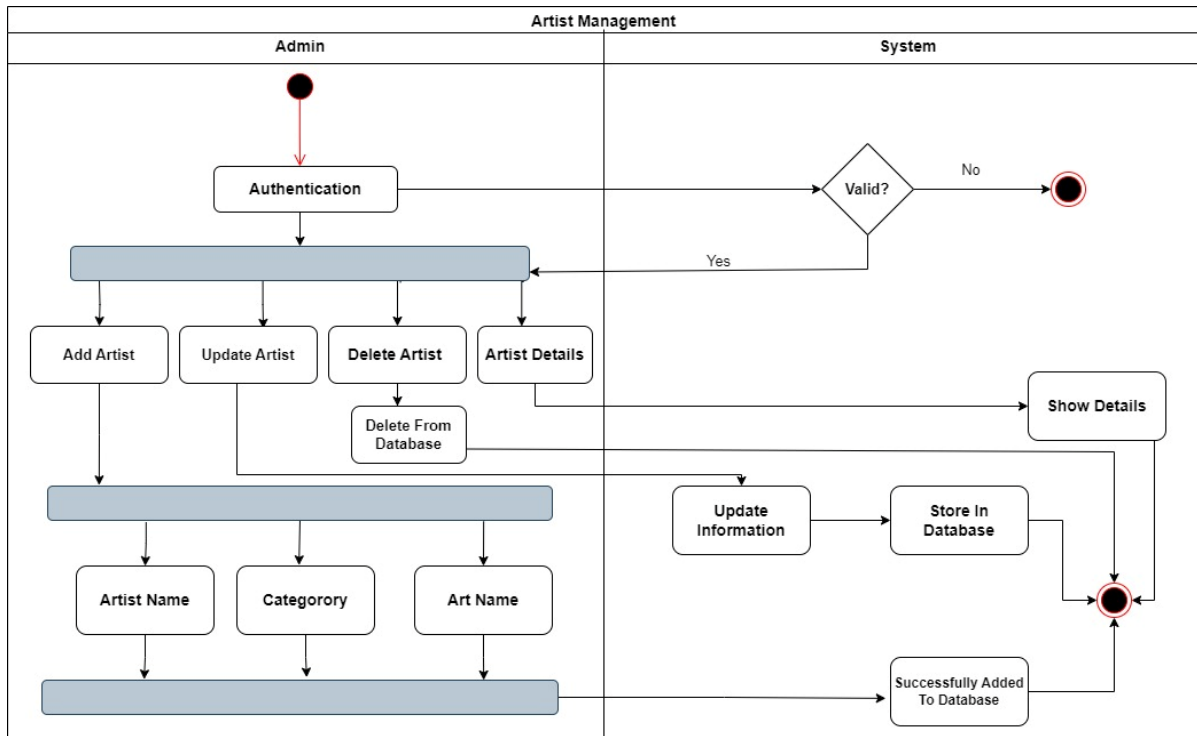
R3: The system acts accordingly and saves the changes.

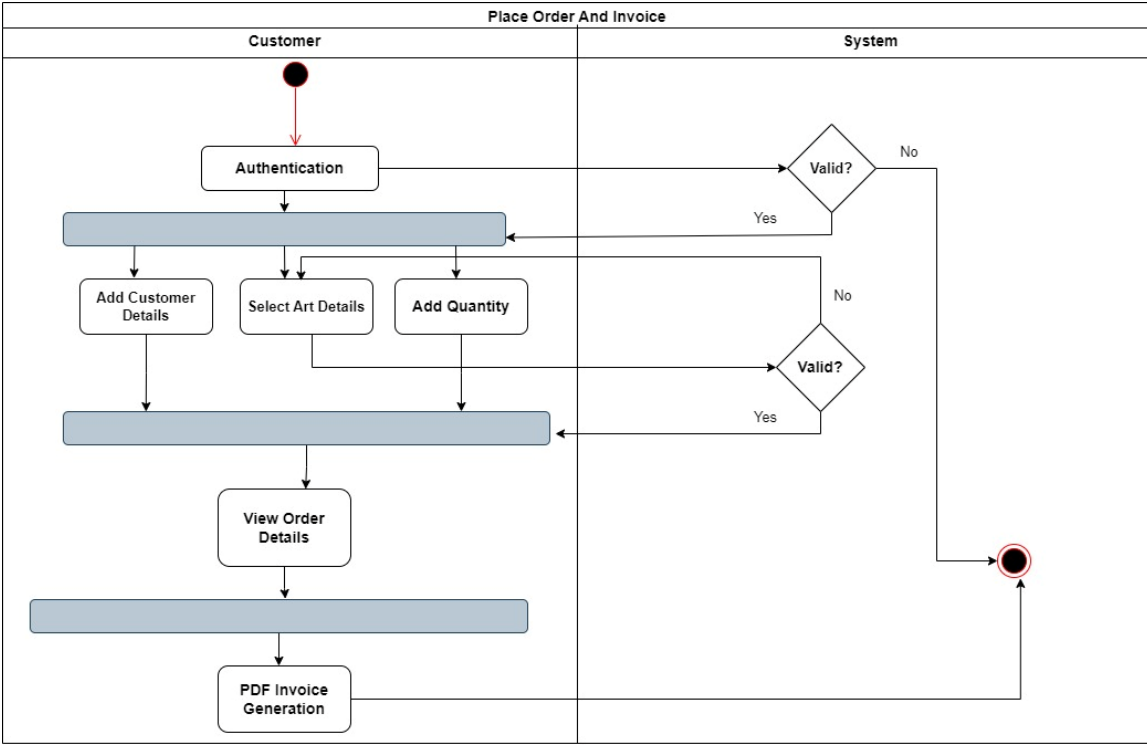
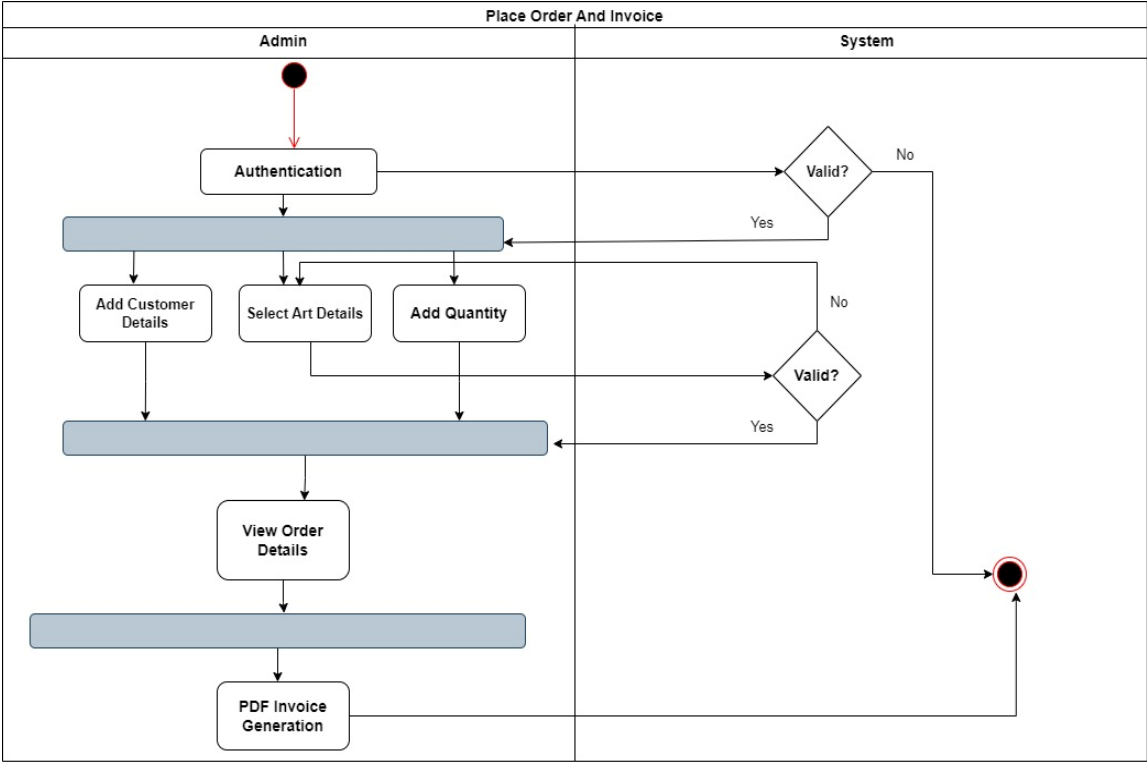
A4: User (customer) selects updates bidding price options.

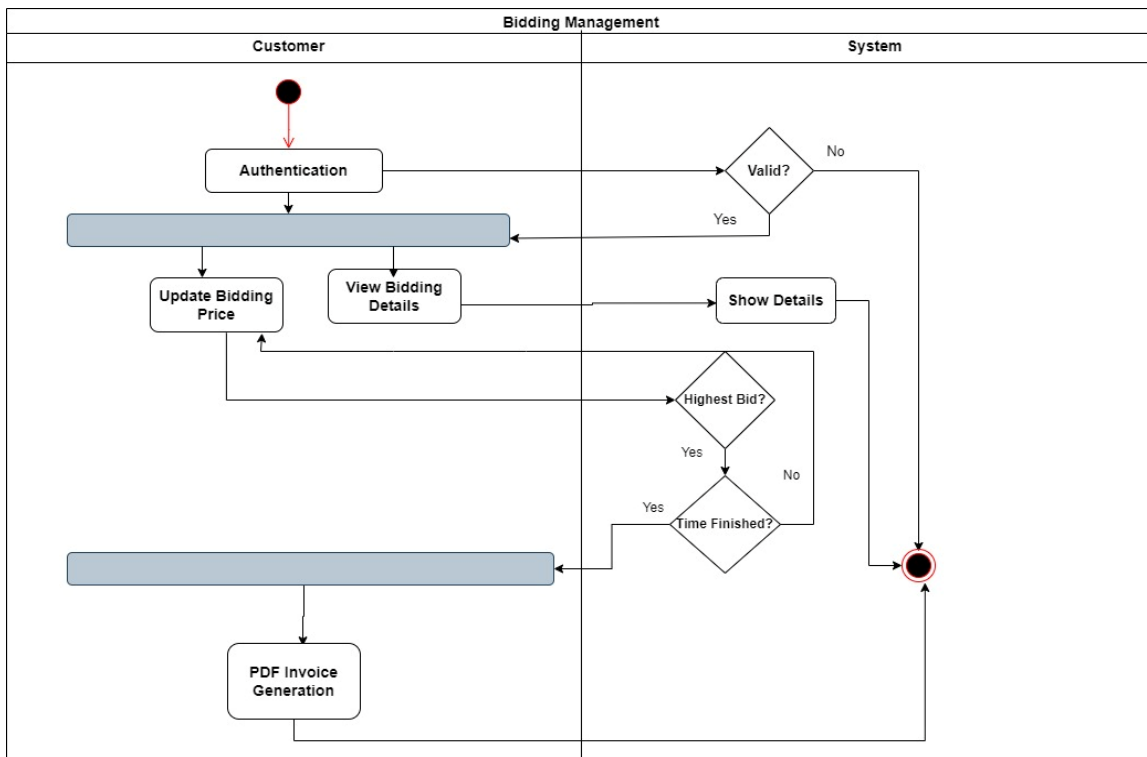
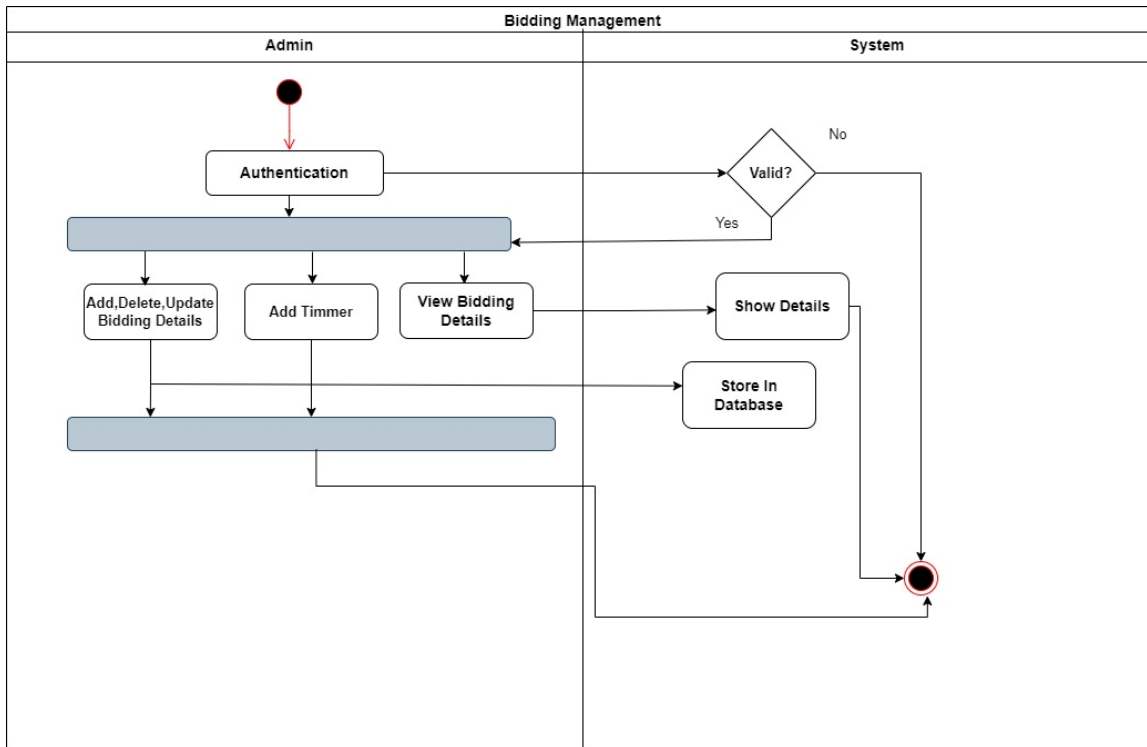
R4: The system acts accordingly and saves the changes.

Swimlane diagrams:

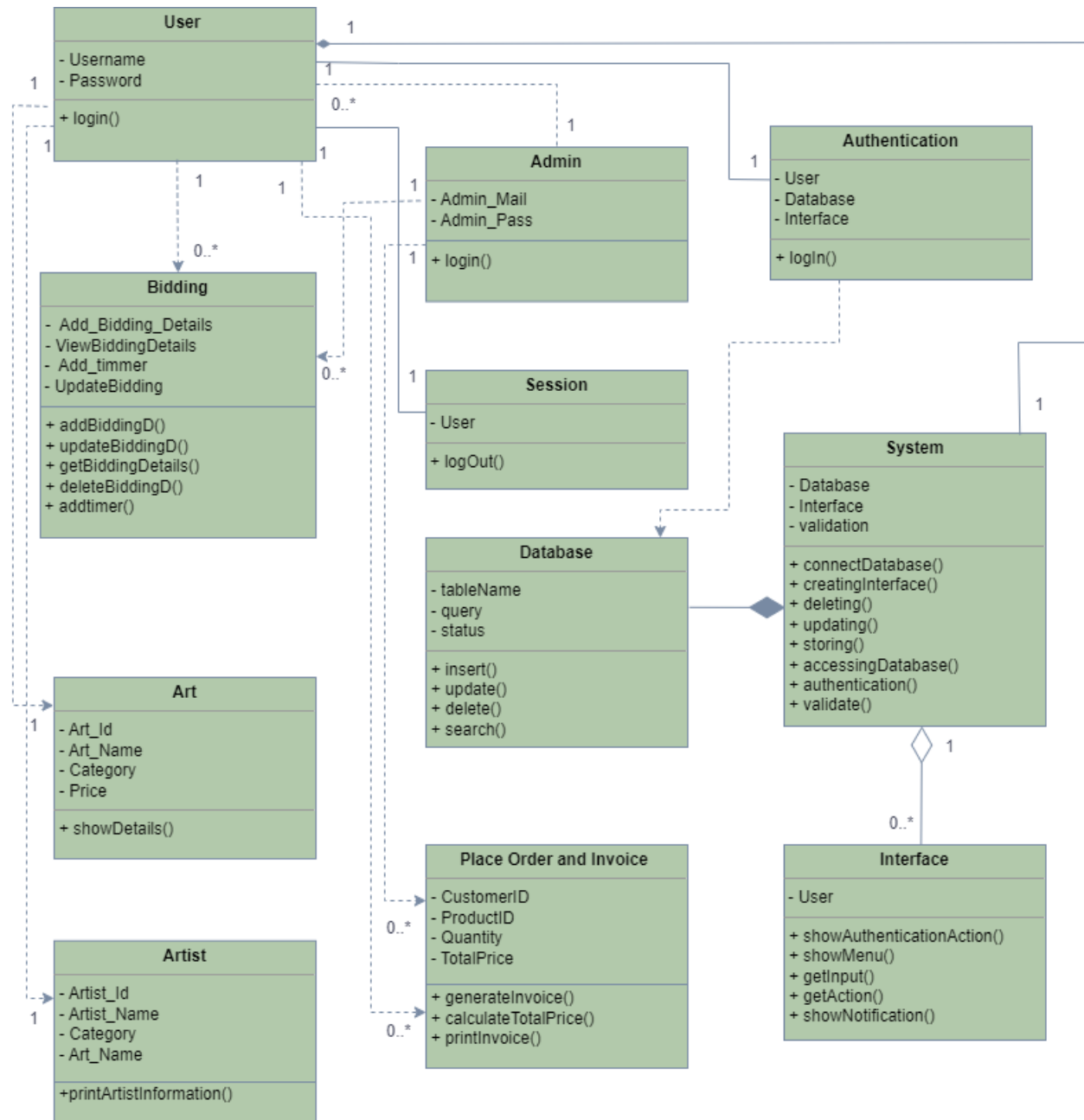




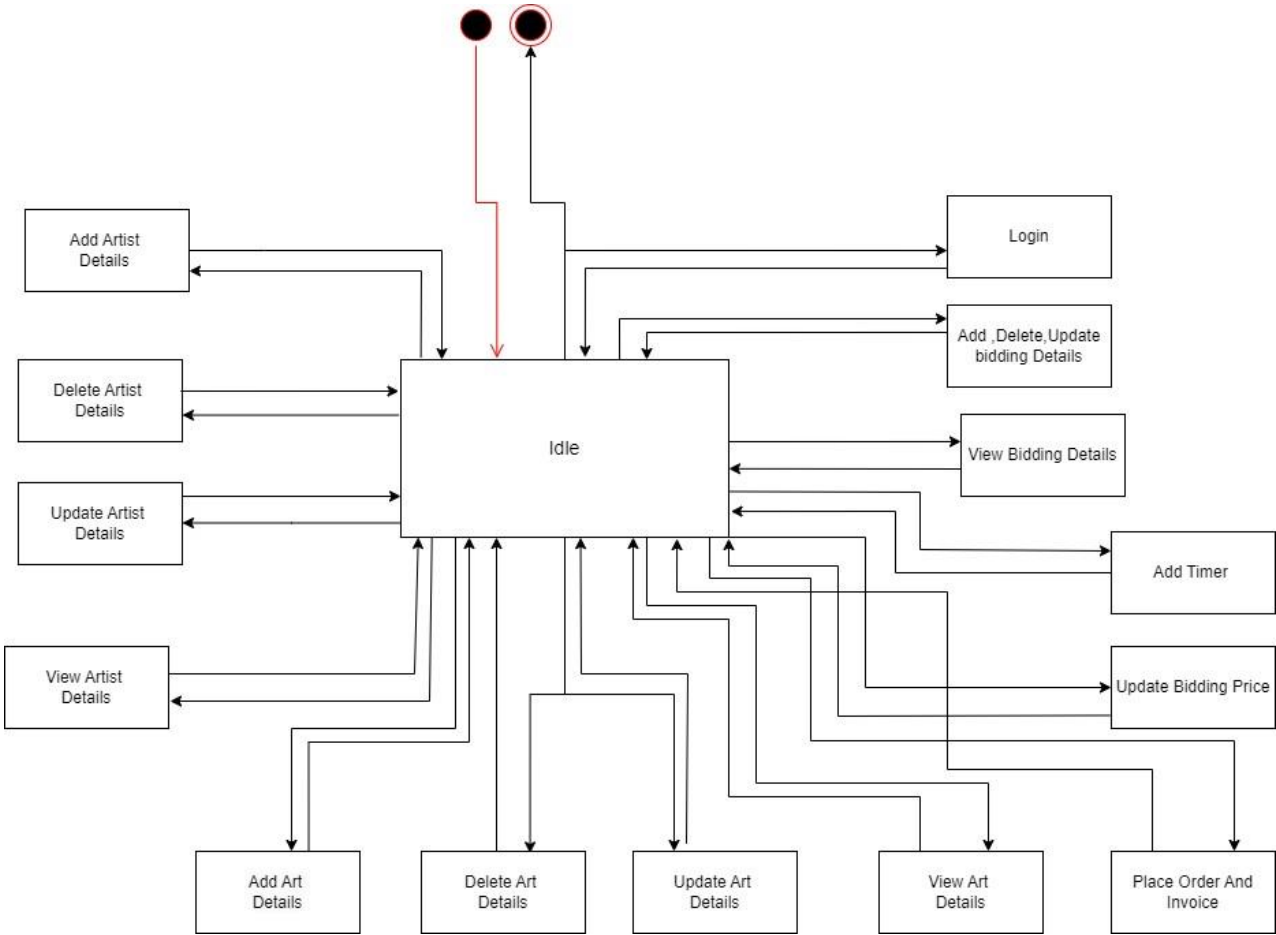




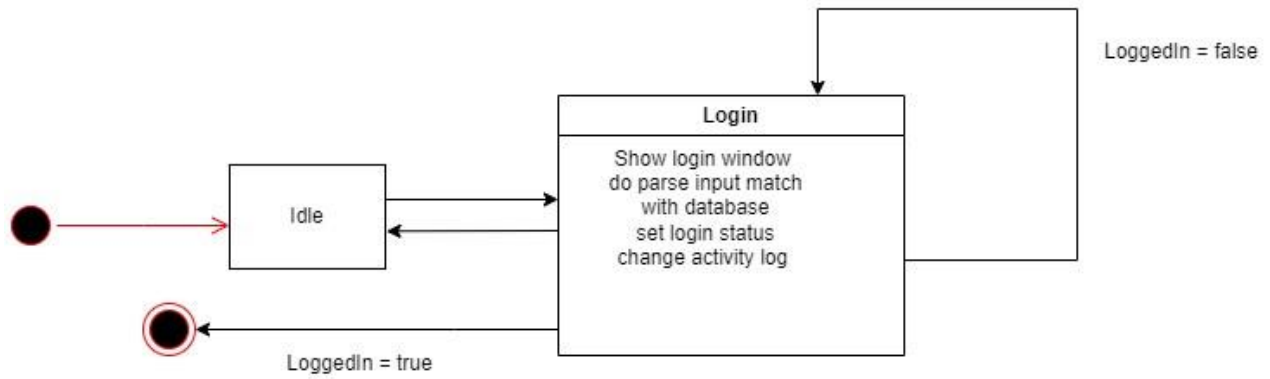
Class diagram:



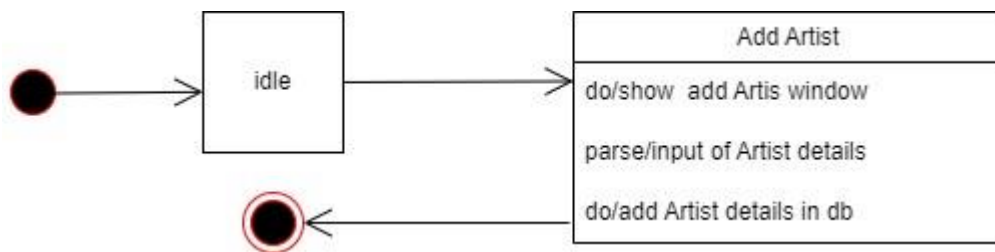
State diagram:



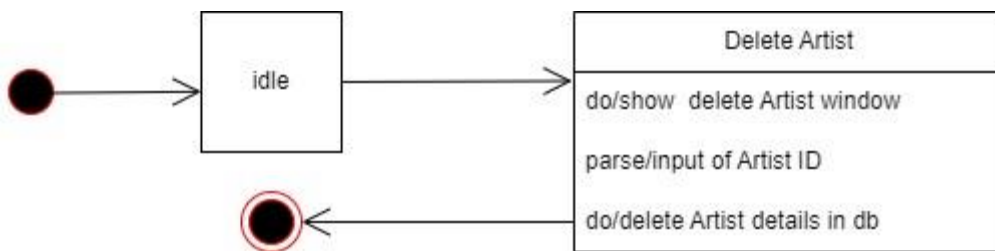
Fig(1)



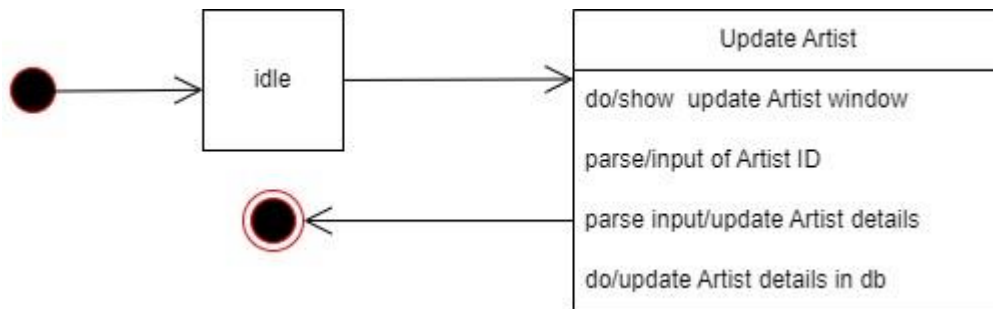
Fig(2)



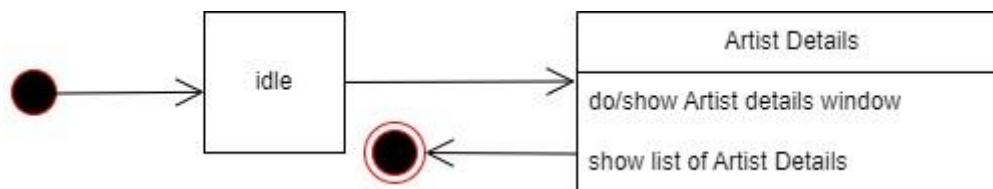
Fig(3)



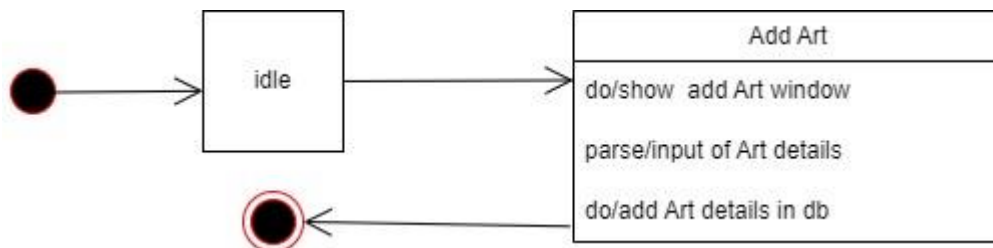
Fig(4)



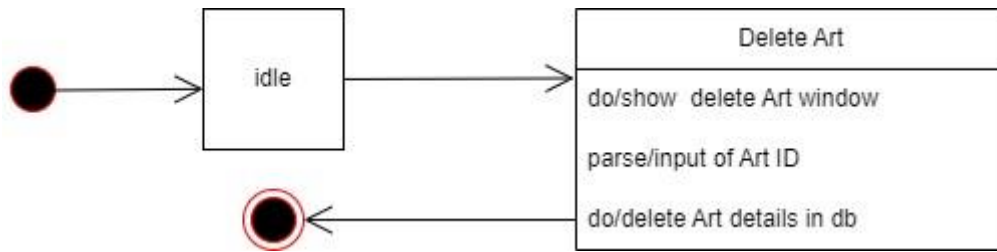
Fig(5)



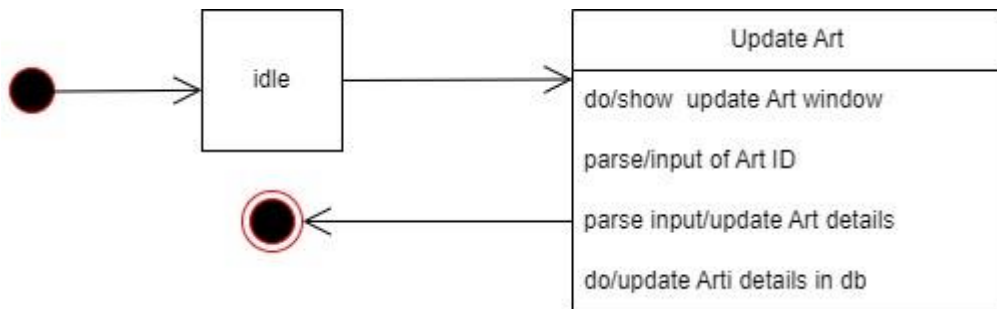
Fig(6)



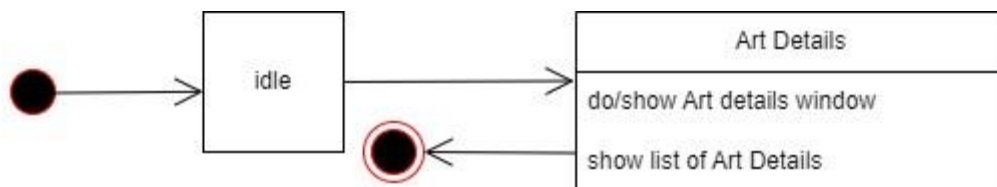
Fig(7)



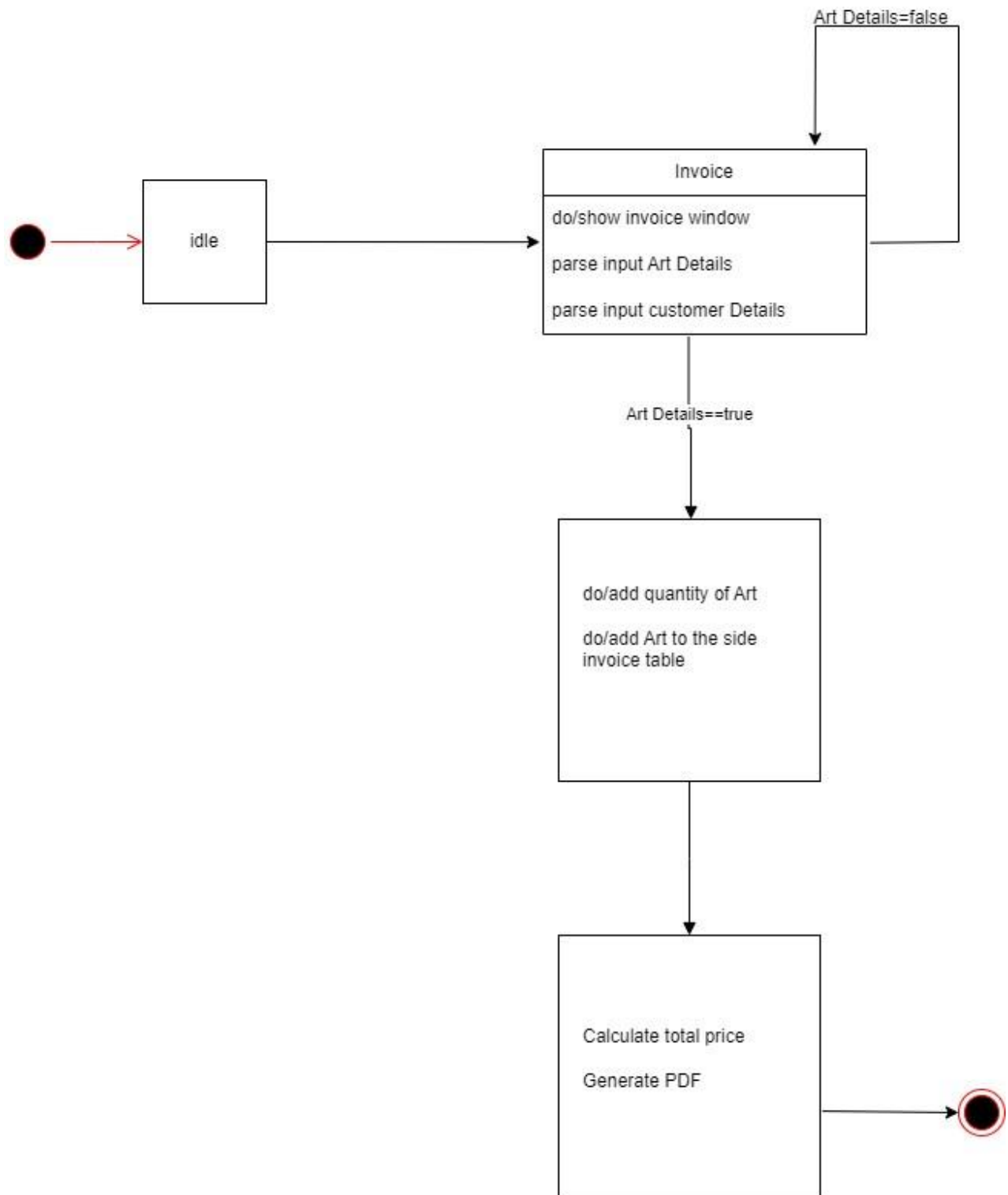
Fig(8)



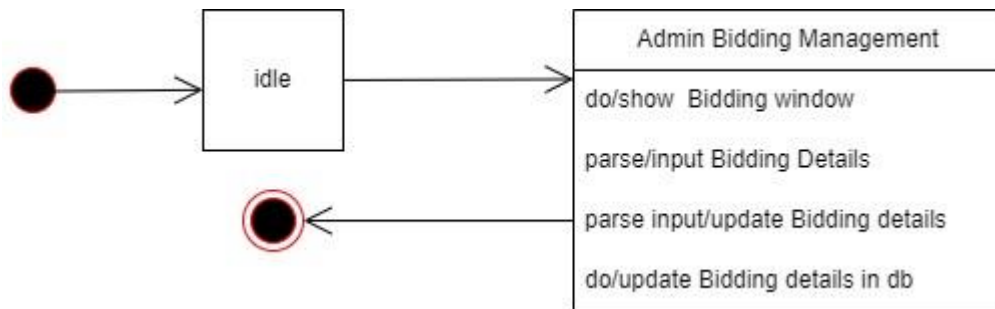
Fig(9)



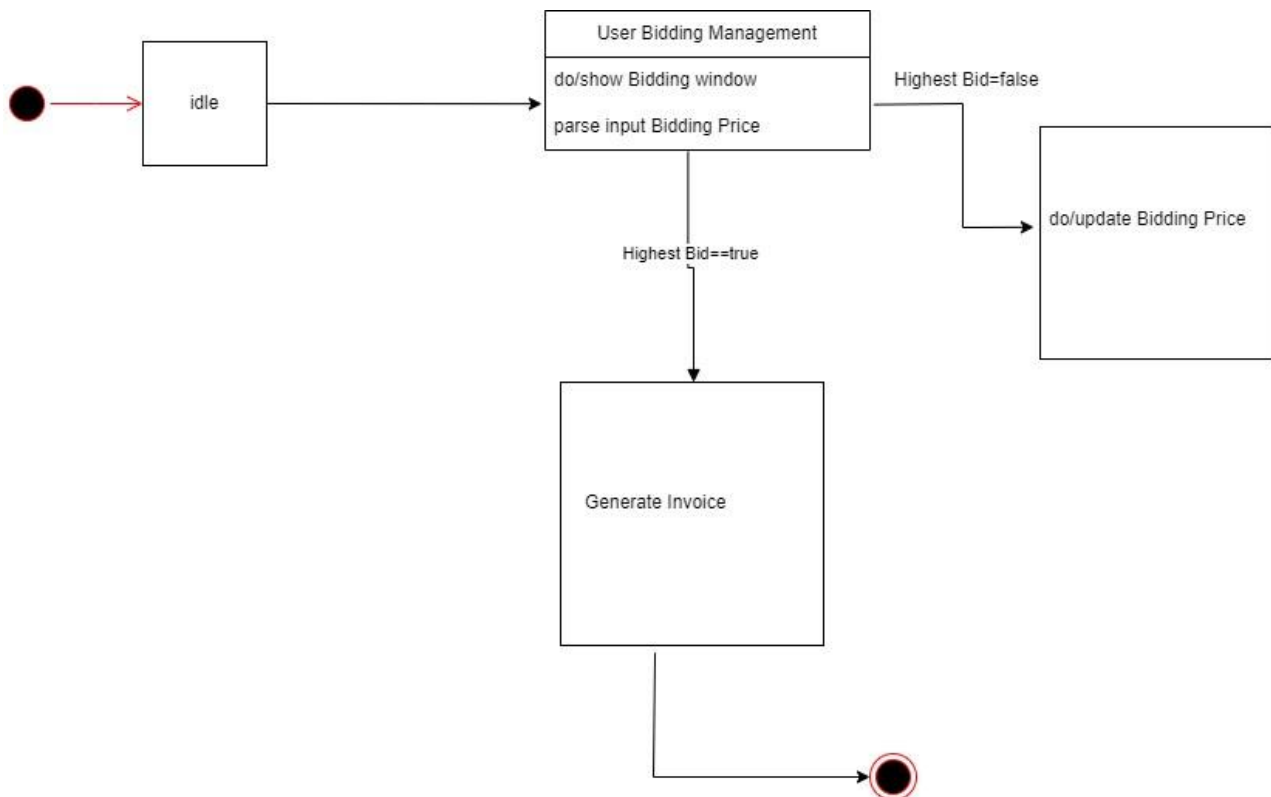
Fig(10)



Fig(11)

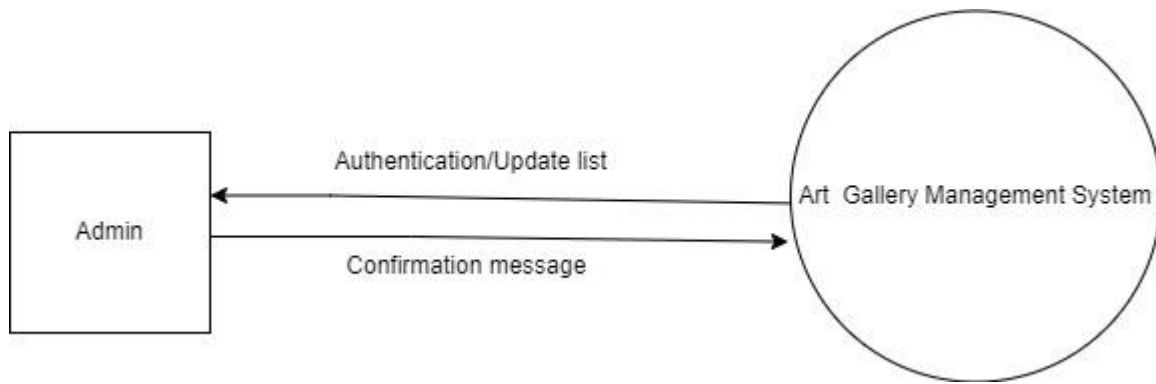


Fig(12)

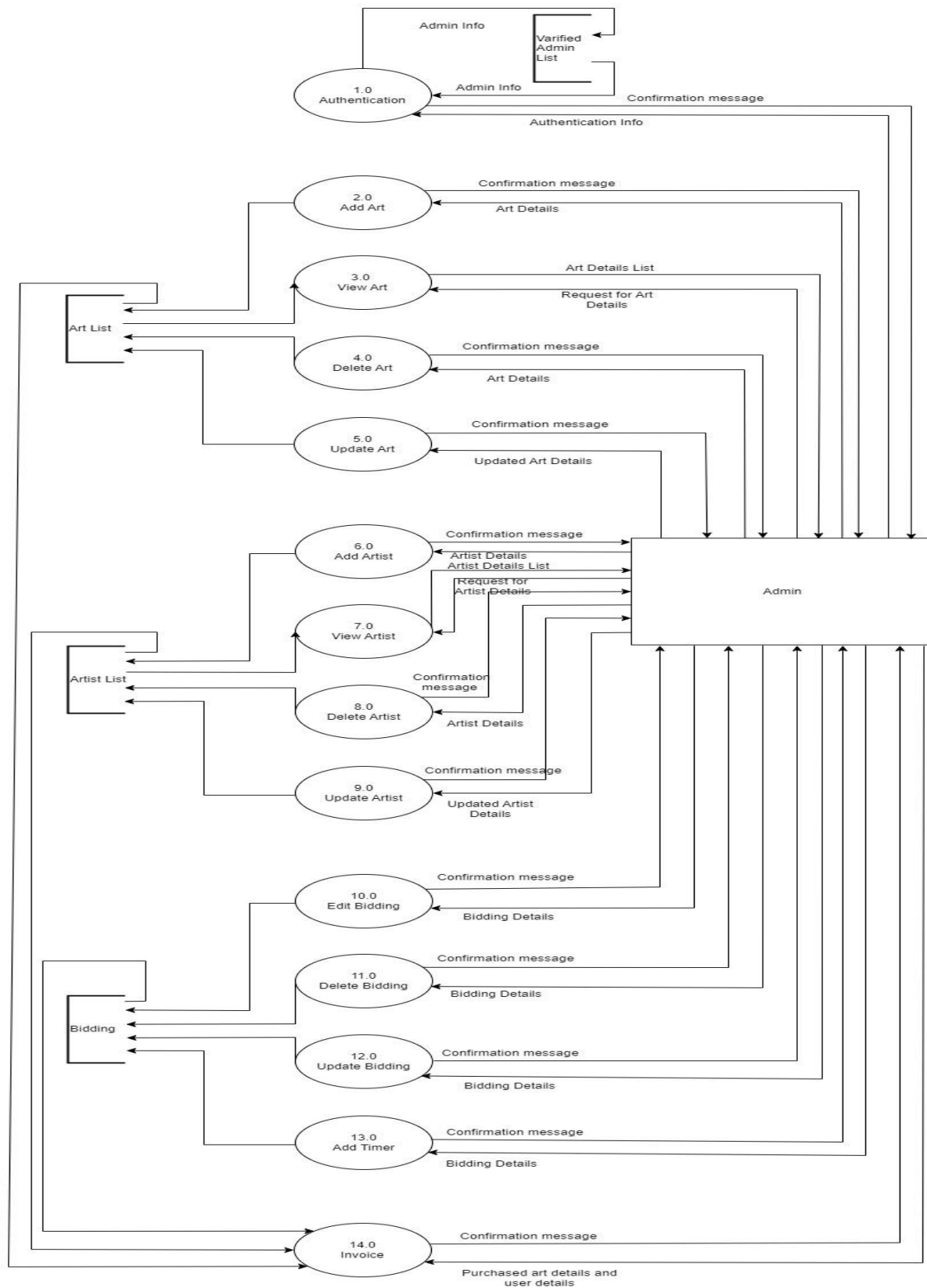


Fig(13)

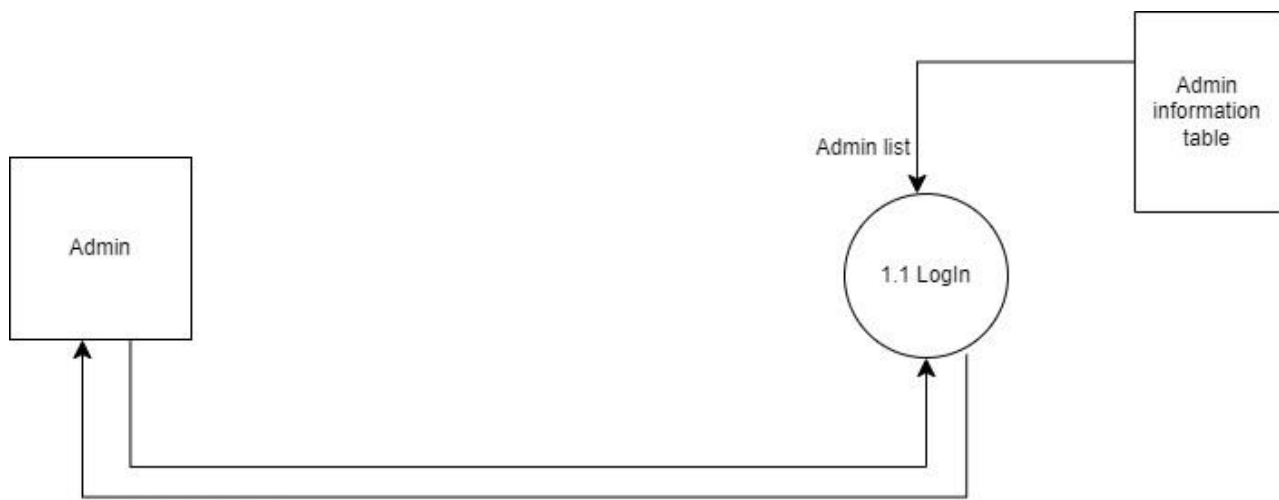
Data flow diagram:



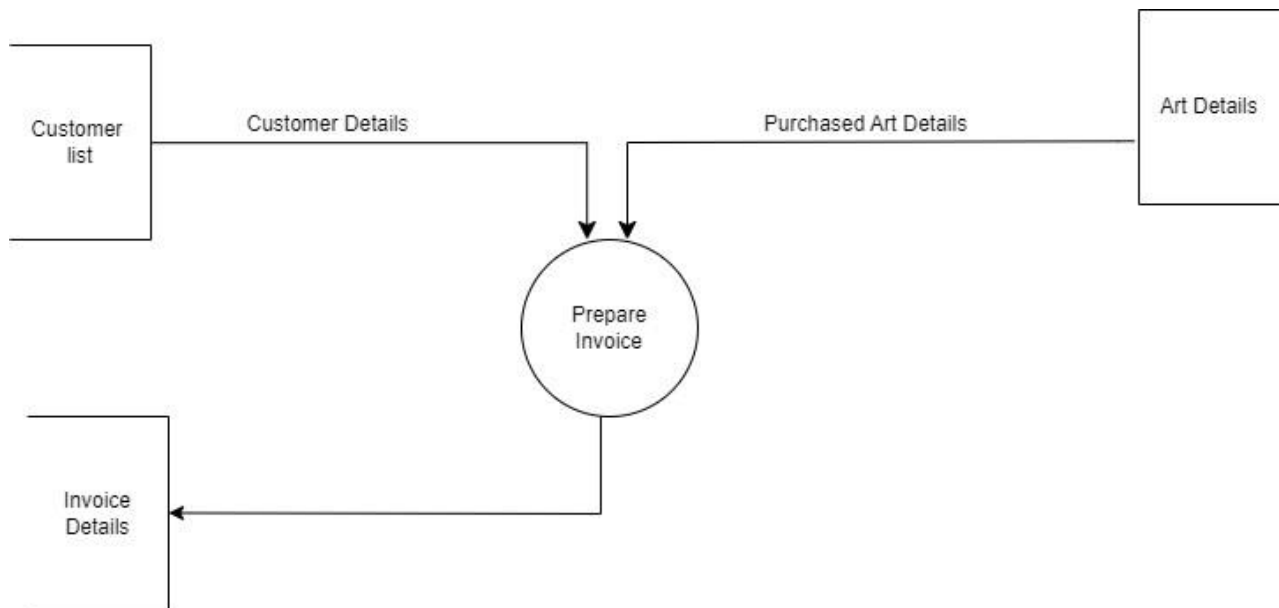
Fig(1)



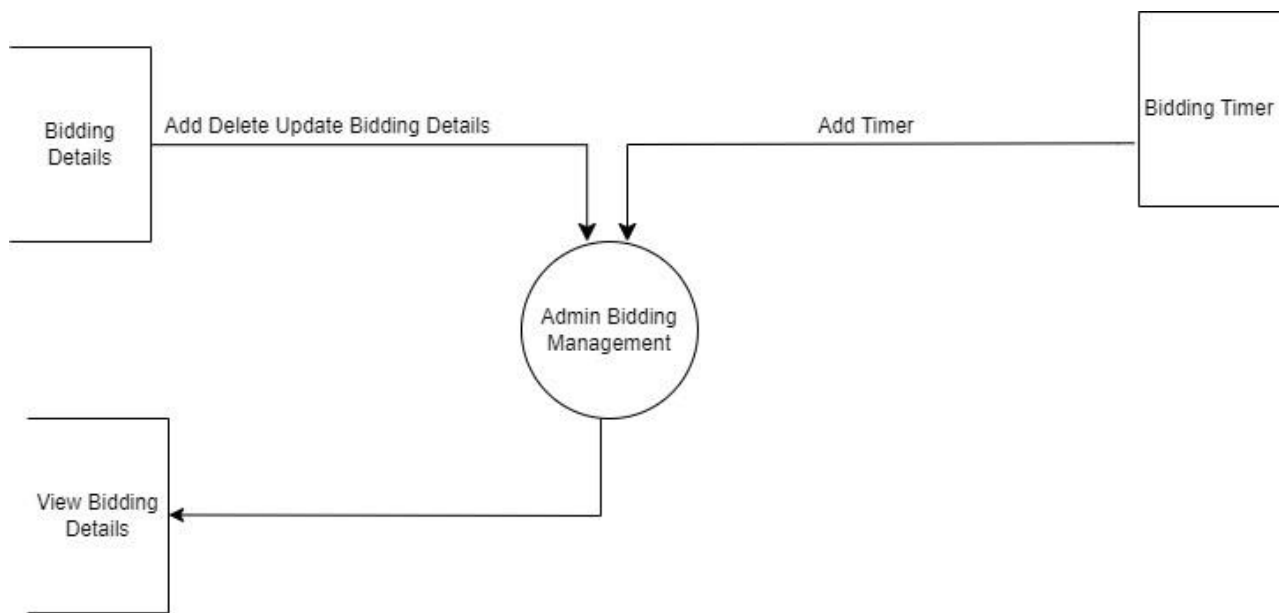
Fig(2)



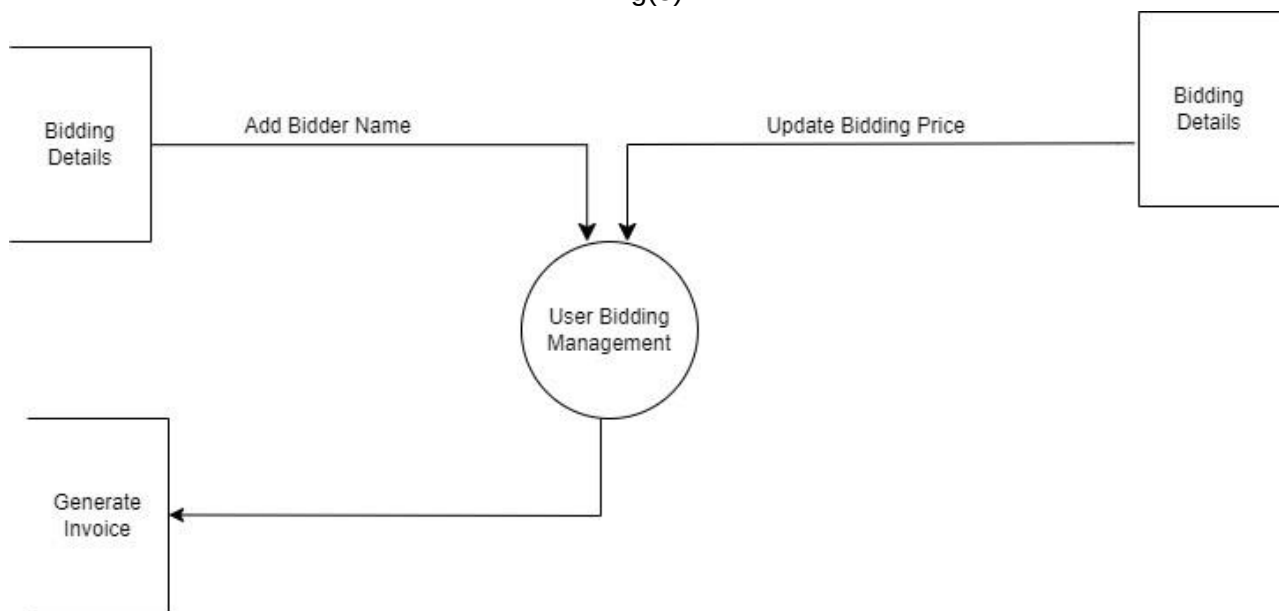
Fig(3)



Fig(4)

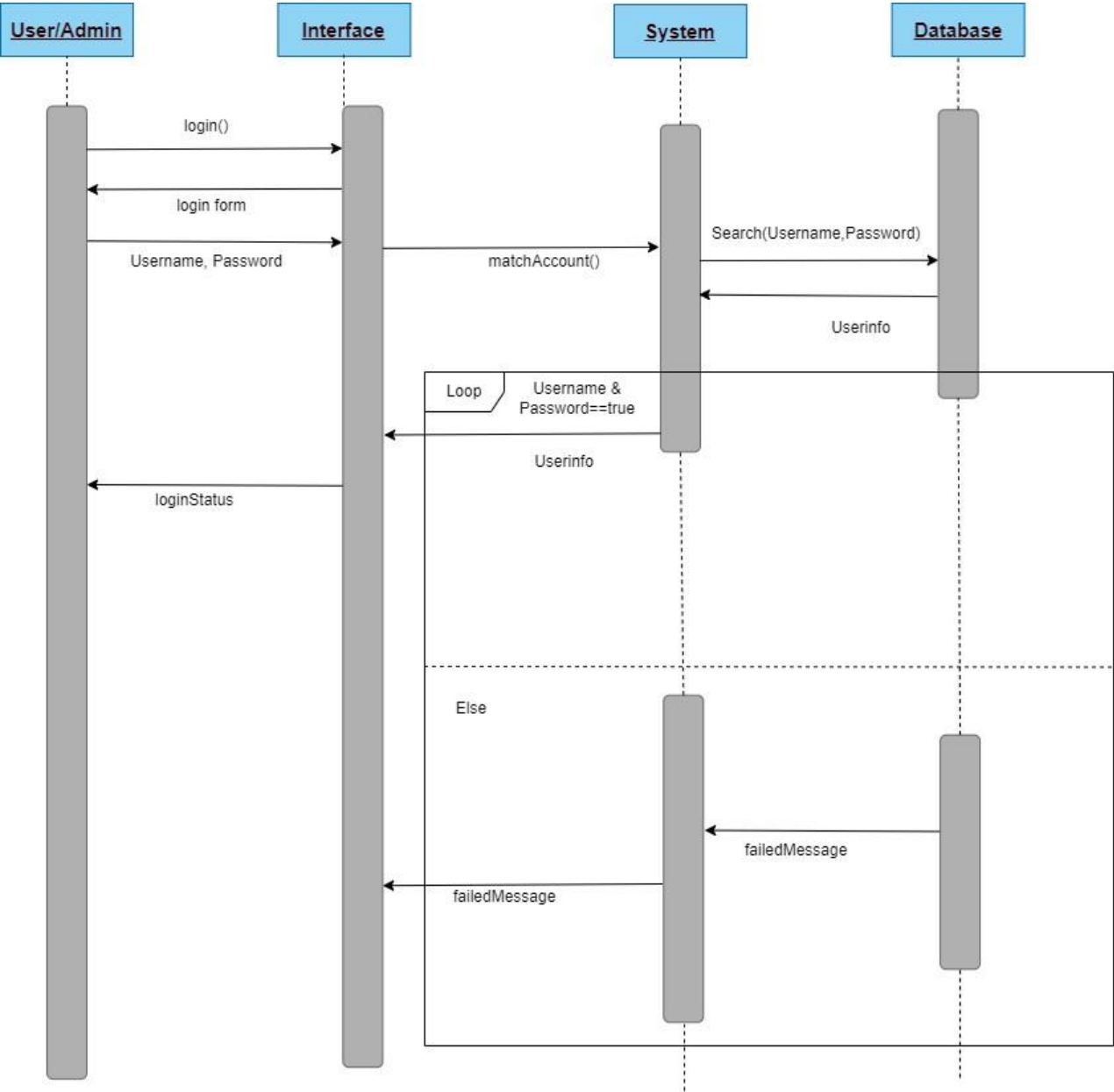


Fig(5)

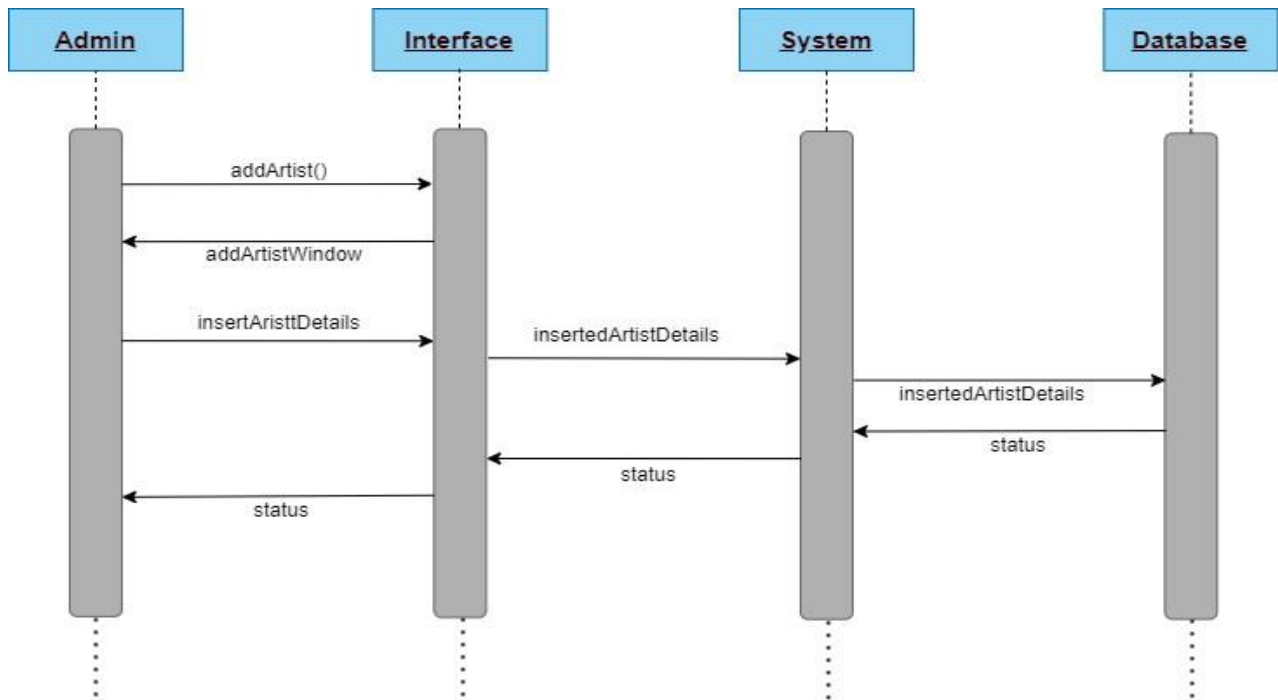


Fig(6)

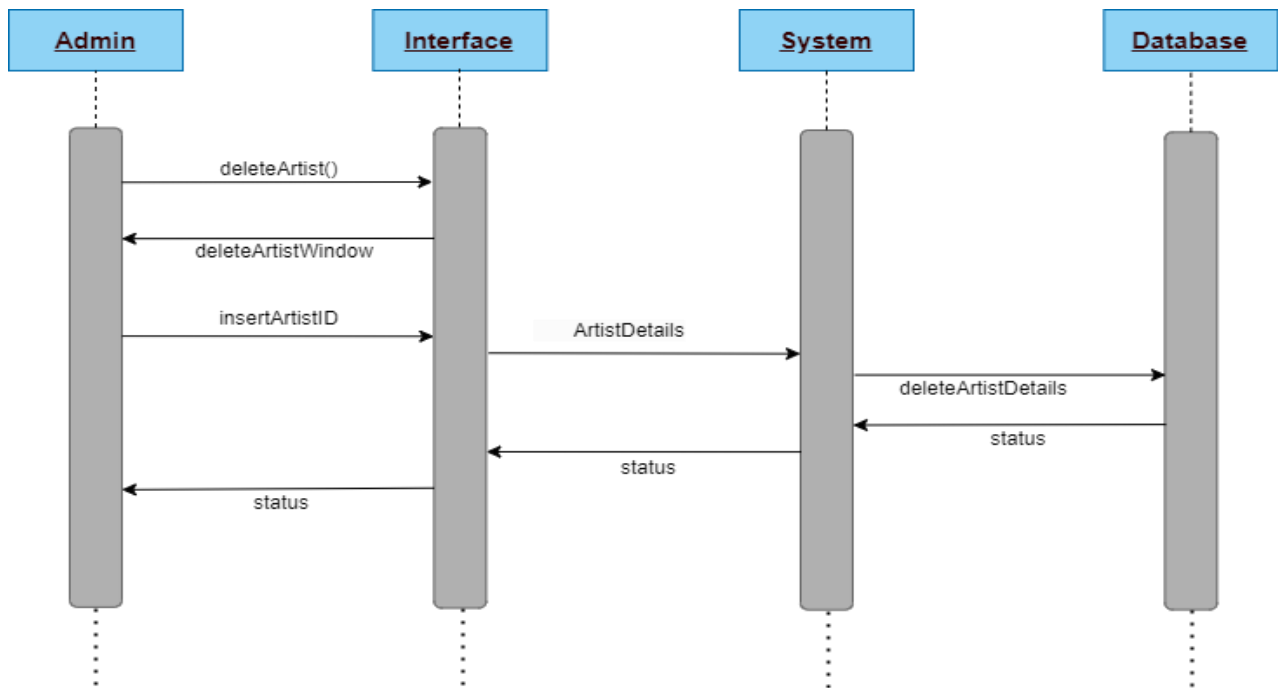
Sequence diagram:



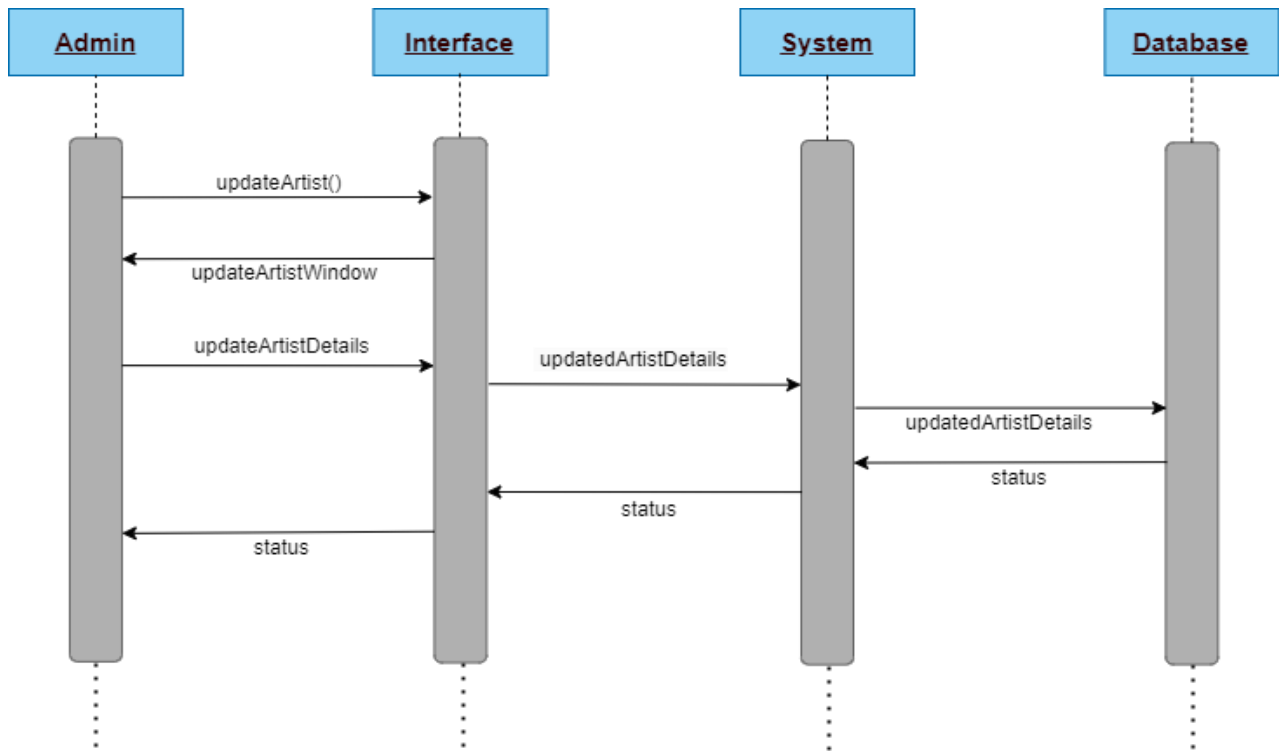
Fig(1)



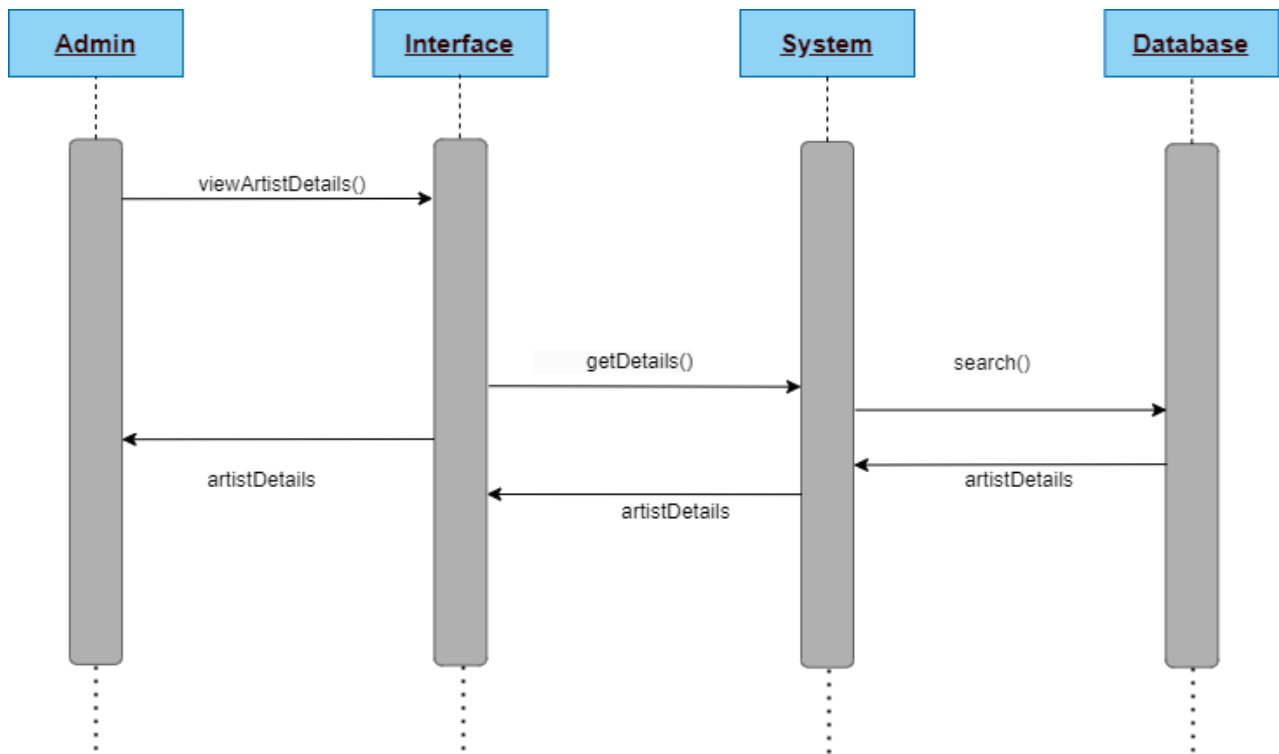
Fig(2)



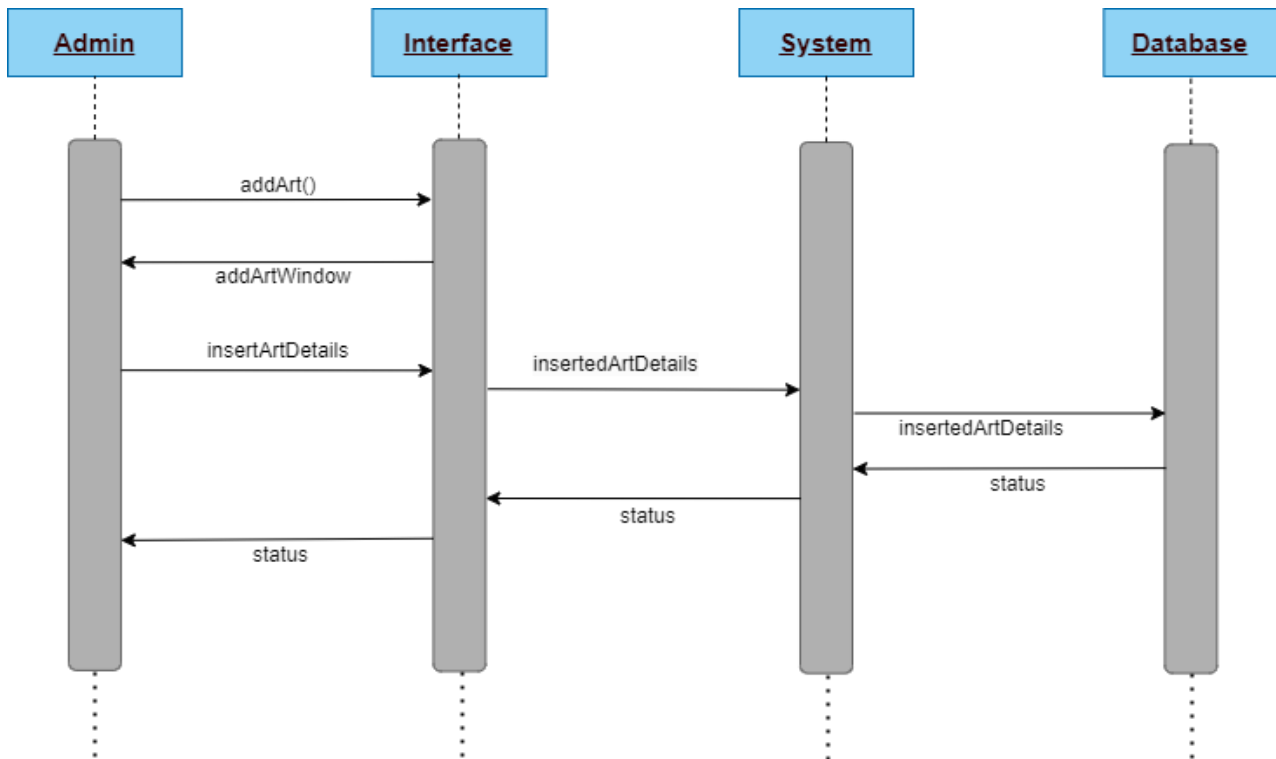
Fig(3)



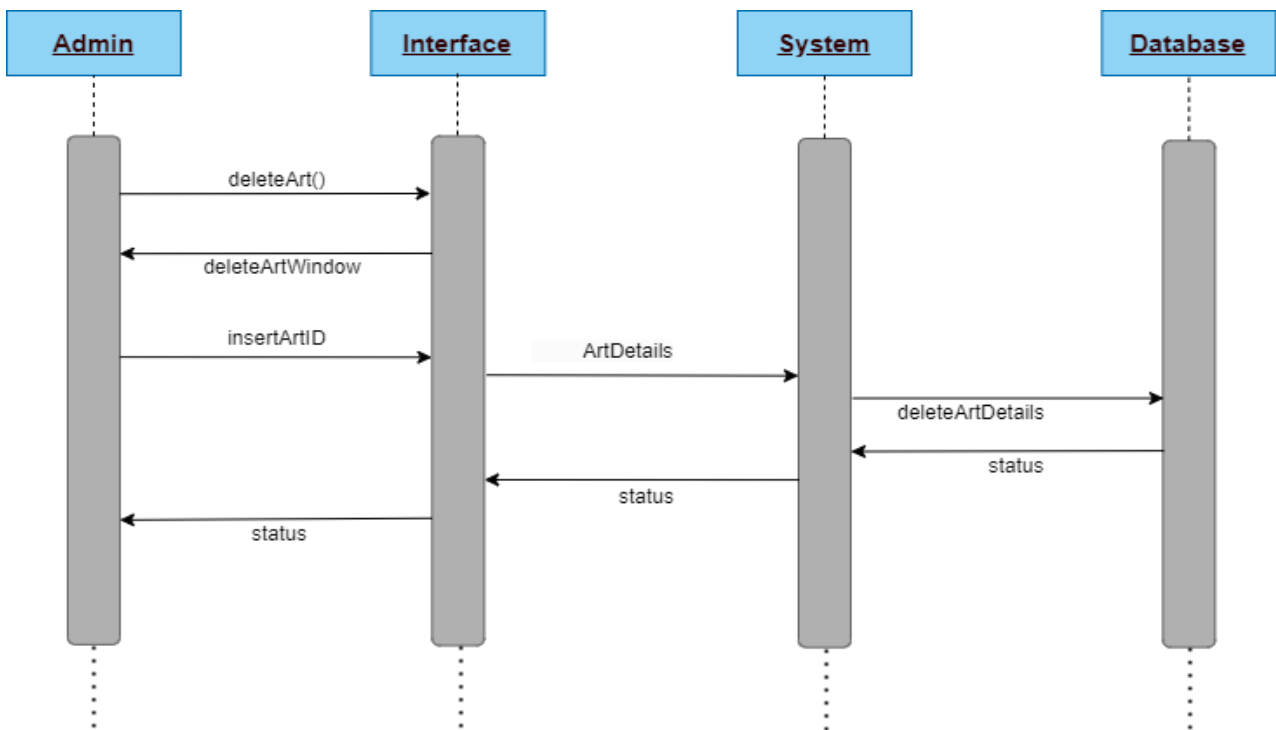
Fig(4)



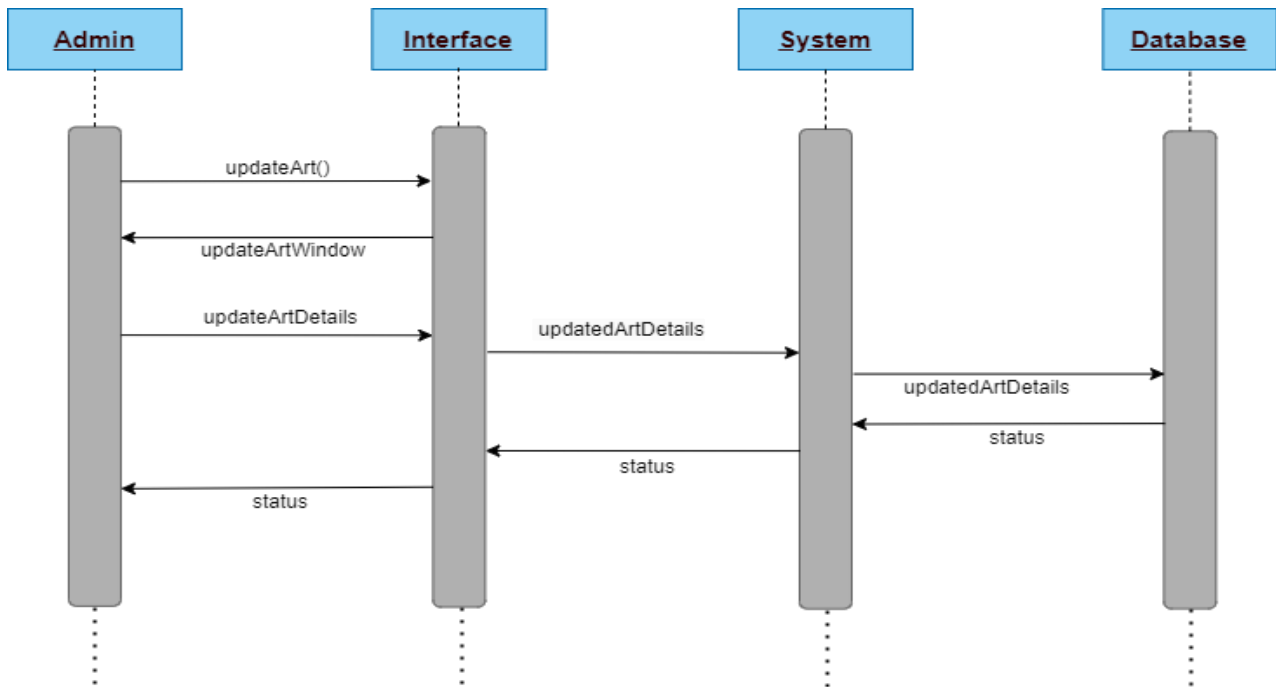
Fig(5)



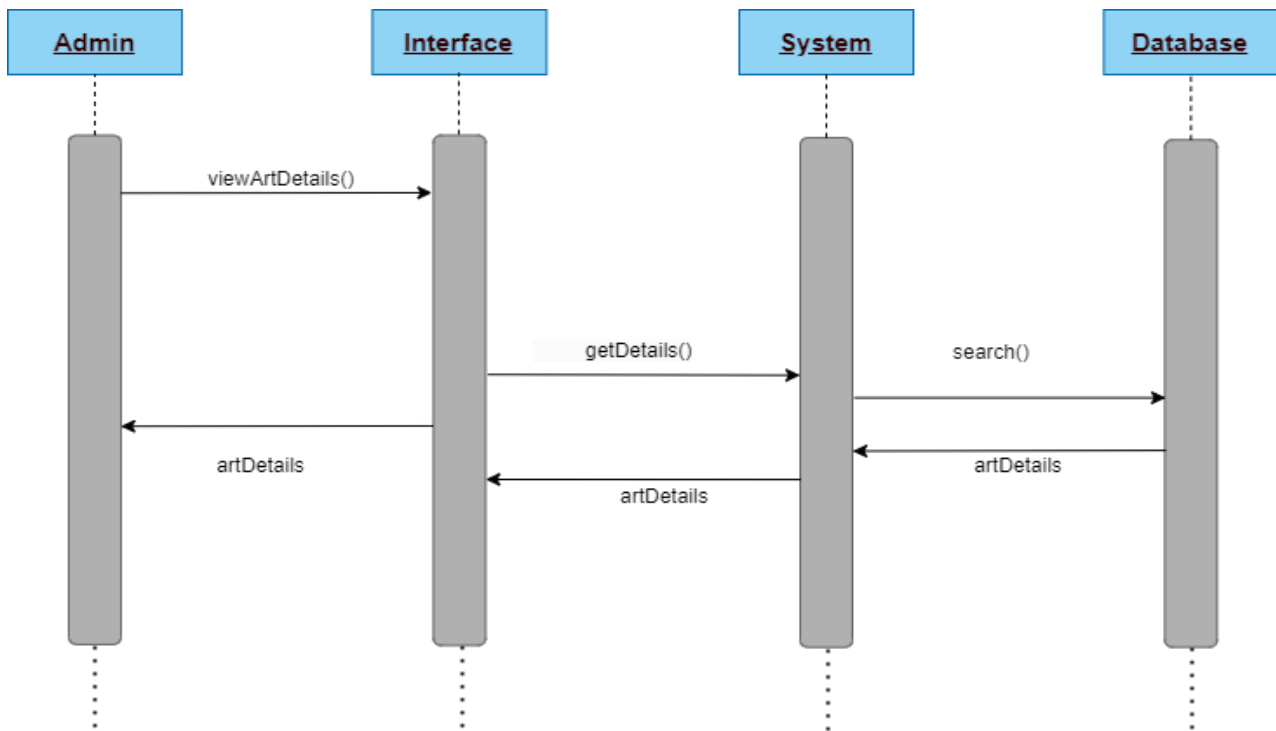
Fig(6)



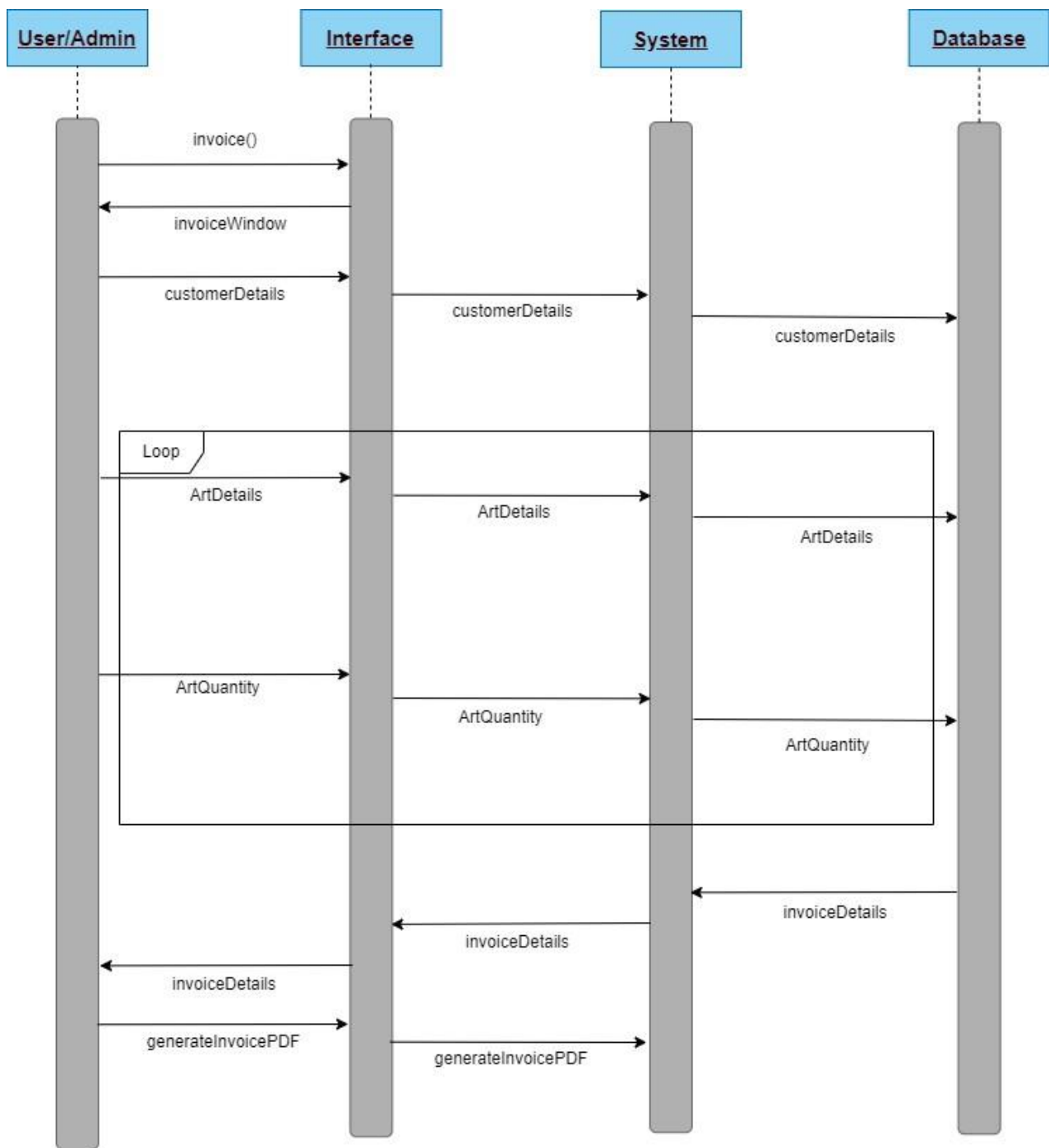
Fig(7)



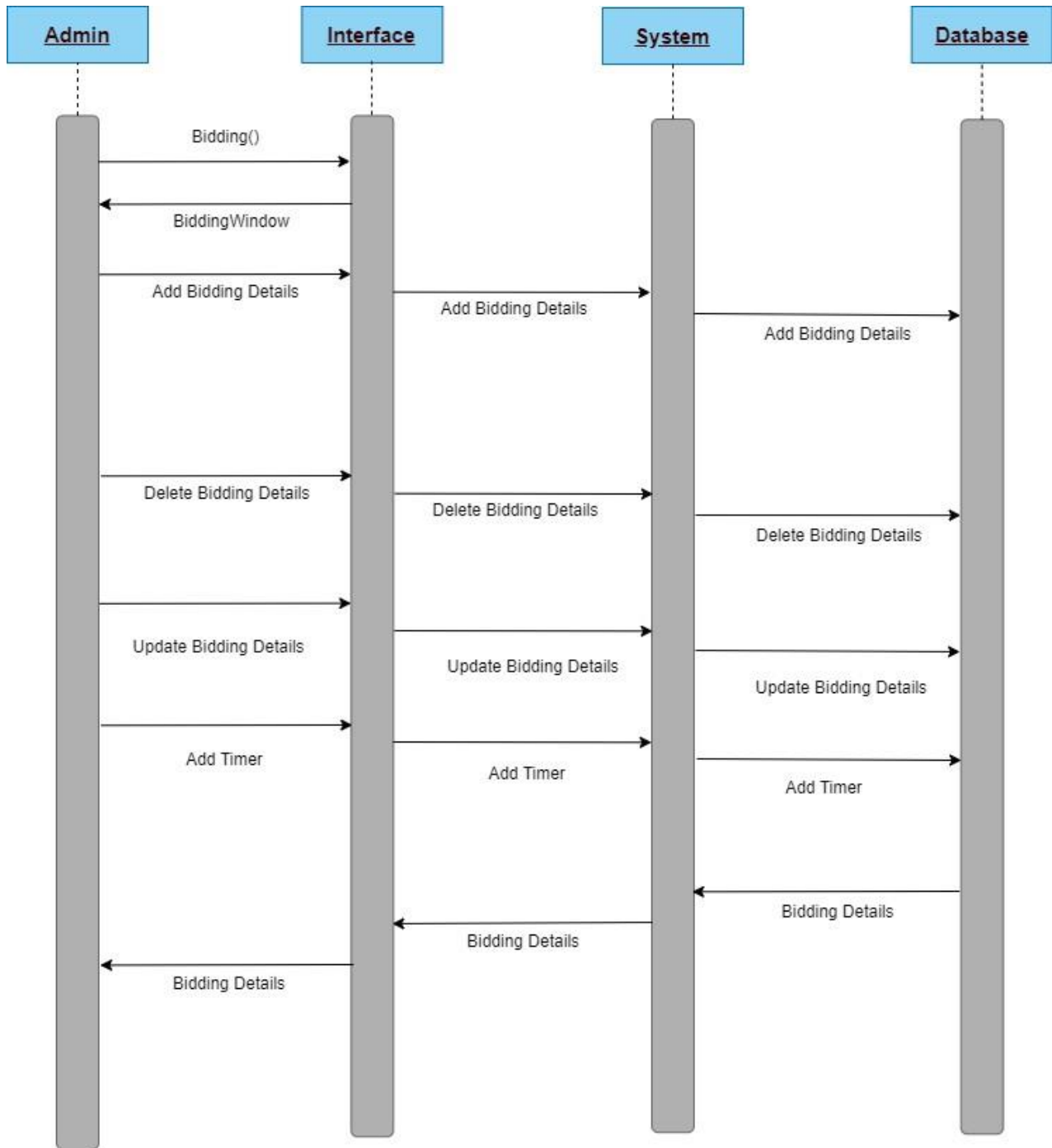
Fig(8)



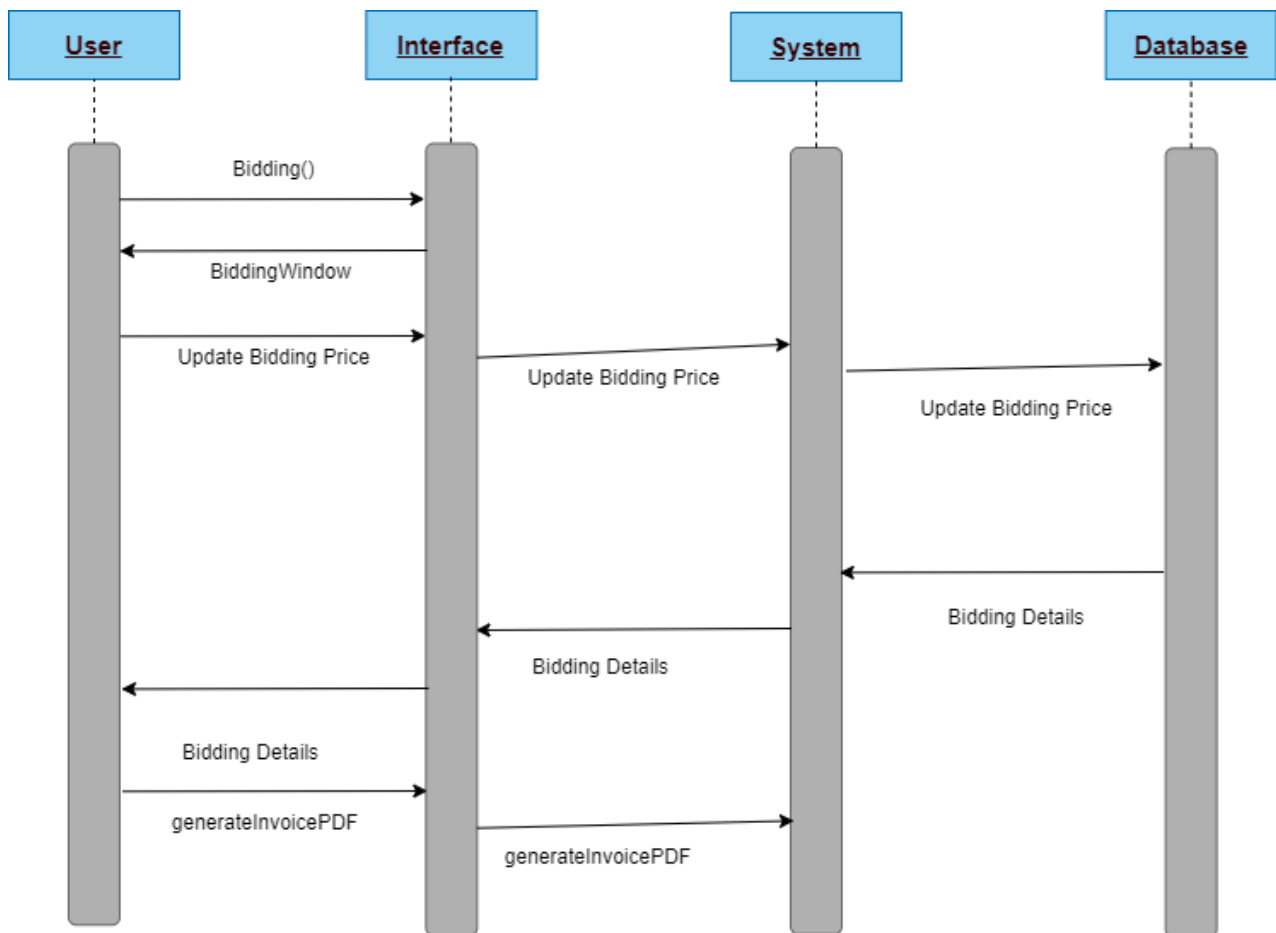
Fig(9)



Fig(10)



Fig(11)



Fig(12)